# Borrowing Strength Without Borrowing Data: Online Supplement

Simulation study, real hospital networks, pre-registrations and the development of GHB

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This supplement accompanies the paper *Borrowing Strength Without Borrowing Data: Private, Serverless and Tuning-Free Collaboration with Graph Hierarchical Borrowing* and follows its order. Section S1 gives the full design and results of the eighteen Monte Carlo simulations of Section 6, which run GHB-lite, with GHB added where the paper reports it. Section S2 documents the real-data study of Section 7.1: data, protocol, the configurations frozen for every literature method, the development results, the ablation of GHB, the oracle arm and the tuning of GHB. Section S3 records the four pre-registrations and their outcomes. Section S4 gives every network's results for the eight-network benchmark of Section 7.2, and Section S5 the design and every hospital's results on UCI Heart Disease. Section S6 documents how GHB-lite was revised after a first run of Simulation 15 and the extensions that were tried and not adopted. Every number is generated by the scripts released with the paper and read into the text as LaTeX macros.

## S1 SIMULATION STUDY

Section 6 of the paper summarizes these simulations; this section gives the full design and every figure. The simulations test GHB-lite (Sections 4 and 5 of the paper) under its own model and under a logistic analogue. They check internal consistency and expose the cost of estimating the unknown quantities that Propositions 4.1 to 4.3 take as known; they are not evidence about real hospital networks. Unless stated otherwise, the network has 24 hospitals whose sizes are drawn once from a log-normal distribution with median 100 and clipped (16 to 479 patients), $\sigma^2 = 1$, $\delta = 10^{-5}$, clipping bound $C = 1$, and the privatized releases are averaged by 60 rounds of Metropolis-Hastings gossip on the ring with two-hop links of Figure 2. *Plug-in GHB-lite* estimates everything it needs from the privatized releases and public quantities only: $\tau^2$ by DerSimonian-Laird [DerSimonian and Laird 1986] (Remark 4.4), the leave-one-out population estimate by inverse-variance weighting, and its error $M$ from those weights. Risk is reported relative to local estimation, $R_k/v_k$ with $v_k = \sigma^2/n_k$; values below 1 mean that a hospital gains. All code is in `simulations/` of the companion survey's repository, and `run_all.py` regenerates every figure and number in about twenty minutes.

*Simulation 1: the closed forms.* Figure S1 and Tables S1 and S2 repeat the setting of Figure 1 with 40,000 replicates per point. Across 101 comparisons of a simulated risk or $M$ with its closed form, 98% lie within two Monte Carlo standard errors, and the largest relative deviation is 1.6%. Propositions 4.1 and 4.2 and the code that draws Figure 1 agree.

*Simulation 2: heterogeneity.* Figure S2 and Table S3 vary $\tau^2$ from $10^{-4}$ to 1. Plug-in GHB-lite is never worse than local estimation by more than Monte Carlo error: its worst hospital reaches a relative risk of 1.02 at most. A single global model (the sample-size weighted mean of all site estimates, computed without privacy) is the best choice when $\tau^2$ is tiny, but its mean relative risk reaches 134 and its worst hospital 382 at $\tau^2 = 1$. Estimating $\tau^2$ costs most when heterogeneity is

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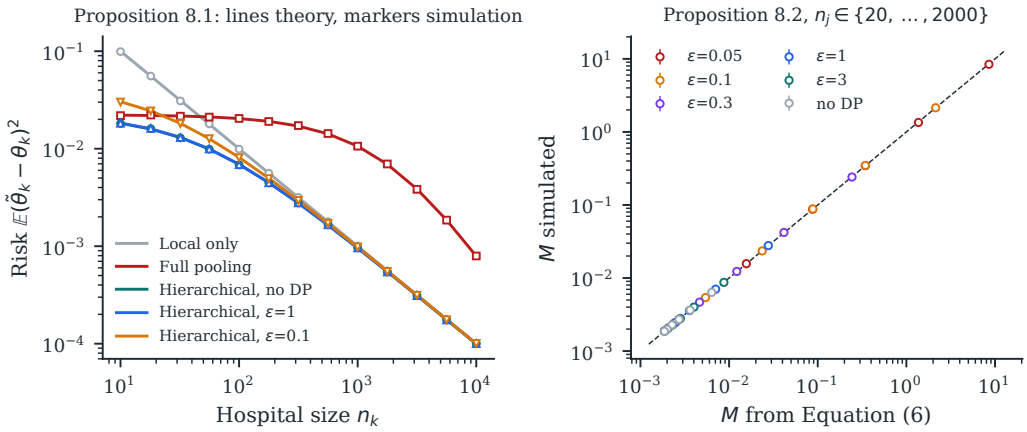



Fig. S1. Simulation 1. Left: risk against hospital size; lines are the closed forms of Figure 1, markers are simulated (±2 standard errors). Right: simulated $M$ of the privatized leave-one-out mean against Equation (4), for 11 other hospitals of 20 to 2,000 patients and six privacy budgets.

Table S1. Simulation 1: exact values behind Figure S1. Top: risk $R_k$ of a hospital of $n_k$ patients ($\sigma^2 = 1$), simulated, with the closed form in grey; Global: full pooling without privacy; Hier.: hierarchical borrowing with known $\tau^2$. Bottom: simulated $M$ of the privatized leave-one-out mean, when the 23 other hospitals all have $n_j$ patients, with Equation (4) in grey.

| $n_k$ | Local | Global | Hier., no DP | Hier., $\epsilon = 1$ | Hier., $\epsilon = 0.1$ |
|---|---|---|---|---|---|
| 10 | 0.0991 (0.1000) | 0.0218 (0.0221) | **0.0181** (0.0182) | 0.0183 (0.0184) | 0.0301 (0.0304) |
| 18 | 0.0556 (0.0556) | 0.0222 (0.0219) | **0.0161** (0.0159) | **0.0161** (0.0160) | 0.0244 (0.0244) |
| 32 | 0.0309 (0.0312) | 0.0215 (0.0216) | **0.0129** (0.0130) | 0.0130 (0.0131) | 0.0181 (0.0182) |
| 56 | 0.0180 (0.0179) | 0.0211 (0.0212) | **0.0099** (0.0099) | **0.0099** (0.0100) | 0.0127 (0.0127) |
| 100 | 0.0099 (0.0100) | 0.0204 (0.0204) | **0.0068** (0.0069) | **0.0068** (0.0069) | 0.0081 (0.0081) |
| 178 | 0.0056 (0.0056) | 0.0190 (0.0191) | **0.0045** (0.0045) | **0.0045** (0.0045) | 0.0049 (0.0050) |
| 316 | 0.0032 (0.0032) | 0.0173 (0.0171) | **0.0028** (0.0028) | **0.0028** (0.0028) | 0.0030 (0.0030) |
| 562 | 0.0018 (0.0018) | 0.0143 (0.0142) | **0.0017** (0.0016) | **0.0017** (0.0016) | **0.0017** (0.0017) |
| 1,000 | **0.0010** (0.0010) | 0.0106 (0.0106) | **0.0010** (0.0010) | **0.0010** (0.0010) | **0.0010** (0.0010) |
| 1,778 | 0.0006 (0.0006) | 0.0070 (0.0069) | **0.0005** (0.0005) | **0.0005** (0.0005) | 0.0006 (0.0006) |
| 3,162 | **0.0003** (0.0003) | 0.0038 (0.0039) | **0.0003** (0.0003) | **0.0003** (0.0003) | **0.0003** (0.0003) |
| 5,623 | **0.0002** (0.0002) | 0.0019 (0.0019) | **0.0002** (0.0002) | **0.0002** (0.0002) | **0.0002** (0.0002) |
| 10,000 | **0.0001** (0.0001) | 0.0008 (0.0008) | **0.0001** (0.0001) | **0.0001** (0.0001) | **0.0001** (0.0001) |

Table S2. Simulation 1, continued: $M$ against the size $n_j$ of the other hospitals and the privacy budget; simulated, with Equation (4) in grey.

| $n_j$ | $\epsilon = 0.05$ | $\epsilon = 0.1$ | $\epsilon = 0.3$ | $\epsilon = 1$ | $\epsilon = 3$ | no DP |
|---|---|---|---|---|---|---|
| 20 | 8.42 (8.54) | 2.14 (2.14) | 0.241 (0.243) | 0.0278 (0.0277) | 0.00872 (0.00873) | 0.00636 (0.00636) |
| 50 | 1.35 (1.37) | 0.348 (0.345) | 0.042 (0.0416) | 0.00703 (0.00705) | 0.004 (0.00402) | 0.00363 (0.00364) |
| 100 | 0.345 (0.344) | 0.088 (0.0881) | 0.0123 (0.0122) | 0.00358 (0.00358) | 0.00279 (0.00282) | 0.00272 (0.00273) |
| 200 | 0.0875 (0.0876) | 0.0235 (0.0236) | 0.00467 (0.00464) | 0.00247 (0.00249) | 0.00227 (0.0023) | 0.00228 (0.00227) |
| 500 | 0.0157 (0.0157) | 0.00539 (0.00541) | 0.00239 (0.00238) | 0.00204 (0.00203) | 0.002 (0.002) | 0.00199 (0.002) |
| 2,000 | 0.0027 (0.00272) | 0.00209 (0.00208) | 0.00189 (0.00189) | 0.00188 (0.00187) | 0.00188 (0.00186) | 0.00186 (0.00186) |

smallest: at $\tau^2 = 10^{-4}$ and $\epsilon = 1$ the plug-in reaches 0.12 against 0.07 with the true $\tau^2$. Gossip with 60 rounds is indistinguishable from exact averaging.

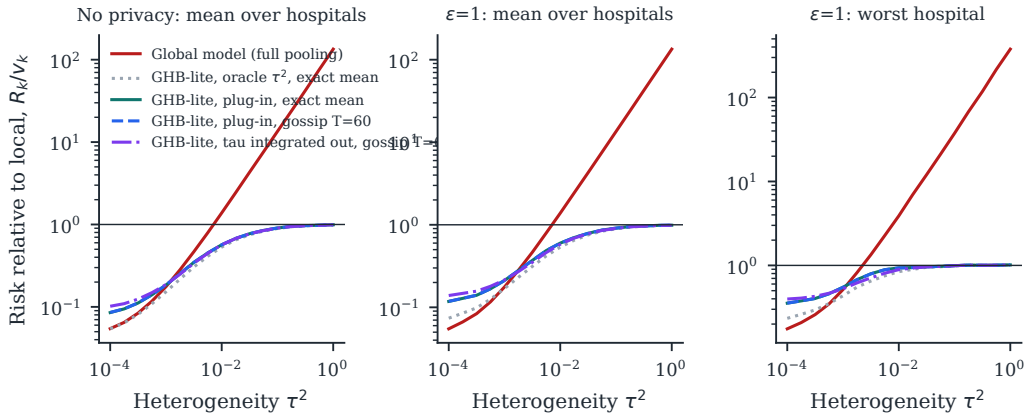



Fig. S2. Simulation 2. Risk relative to local estimation as a function of heterogeneity, mean over hospitals without privacy and with $\epsilon = 1$, and for the worst hospital with $\epsilon = 1$.

*Simulation 3: who gains, and how many hospitals are needed.* At $\tau^2 = 0.02$ and $\epsilon = 1$ (Figure S3, left), every hospital gains about the same number of virtual patients, $B \approx 46$, as Equation (3) predicts. The hospitals with 60 patients or fewer multiply their effective sample size by 2.28 on average, and those with 300 or more by 1.08; the smallest gain of any hospital is 1.05. The global model multiplies the effective sample size of small hospitals by 1.44 but cuts that of large hospitals to 0.15 of their own. With few hospitals, $\tau^2$ is poorly estimated (Figure S3, right, and Table S4). With three hospitals the mean relative risk is 0.91 with DerSimonian-Laird [DerSimonian and Laird 1986] against 0.78 with the true $\tau^2$, and the gap closes as the network grows (0.68 against 0.66 with 48 hospitals). The posterior mean of $\tau^2$ under a half-Cauchy(0.5) prior on $\tau$, the weakly informative choice that Remark 4.4 recommends, does better than DerSimonian-Laird [DerSimonian and Laird 1986] from four hospitals upward but not with three.

*Simulation 4: gossip rounds.* Figure S4 and Table S5 show the mean relative risk of plug-in GHB-lite after $T$ rounds of gossip on the four topologies of Figure 2. The risk comes within 2% of its exact-average value after 35 rounds on the ring, 22 on the regional clusters, 11 on the sparse random graph and 14 on the ring with two-hop links, far fewer than the rounds that the worst-case bound of Proposition 4.3 requires for a consensus error of 1% (203, 127, 73 and 69). The bound is conservative for the risk: it controls the consensus error of the releases, while the risk depends on that error only through a population estimate that the shrinkage weight already discounts. After a single round on the ring, GHB-lite is worse than local estimation (relative risk 1.13), so a hospital should not use the borrowed estimate before the gossip has mixed.

*Simulation 5: joint release by small hospitals.* Section 4.2 proposes that small hospitals release jointly through secure aggregation, so that the privacy noise is calibrated to the group. Figure S5 and Table S6 show that the remedy matters only when every hospital is small. In a network of 16 hospitals of 20 patients and 8 of 200, the large hospitals dominate the inverse-variance weighted population estimate, and groups of eight raise the borrowed patients only from 19.3 to 21.0 at $\epsilon = 0.1$. In a network of 24 hospitals of 20 patients, the same grouping raises $B$ from 1.0 to 6.9 at $\epsilon = 0.1$ and from 30.6 to 41.4 at $\epsilon = 1$.

Table S3. Simulation 2: exact values behind Figure S2. Relative risk $R_k/v_k$, mean over hospitals and worst hospital, at $\epsilon = 1$ (top) and without privacy (bottom). Global: full pooling, without privacy in both blocks; Oracle: GHB-lite with the true $\tau^2$ and exact averaging; Plug-in: DerSimonian-Laird [DerSimonian and Laird 1986], with exact averaging or 60 gossip rounds; FB: $\tau$ integrated out (Remark 4.4), 60 gossip rounds.

| | Global | | Oracle | | Plug-in, exact | | Plug-in, gossip | | FB, gossip | |
|---|---|---|---|---|---|---|---|---|---|---|
| $\tau^2$ | mean | worst | mean | worst | mean | worst | mean | worst | mean | worst |
| 0.0001 | **0.055** | **0.18** | 0.074 | 0.23 | 0.118 | 0.35 | 0.118 | 0.36 | 0.139 | 0.40 |
| 0.00018 | **0.066** | **0.21** | 0.085 | 0.26 | 0.128 | 0.38 | 0.129 | 0.38 | 0.148 | 0.41 |
| 0.00032 | **0.084** | **0.26** | 0.099 | 0.30 | 0.140 | 0.40 | 0.140 | 0.40 | 0.157 | 0.43 |
| 0.00056 | **0.117** | **0.35** | 0.126 | **0.35** | 0.168 | 0.46 | 0.168 | 0.46 | 0.180 | 0.47 |
| 0.001 | 0.176 | 0.52 | **0.169** | **0.43** | 0.212 | 0.55 | 0.212 | 0.55 | 0.215 | 0.53 |
| 0.0018 | 0.282 | 0.82 | **0.231** | **0.55** | 0.279 | 0.68 | 0.279 | 0.68 | 0.271 | 0.62 |
| 0.0032 | 0.465 | 1.34 | **0.313** | **0.65** | 0.370 | 0.80 | 0.370 | 0.79 | 0.351 | 0.71 |
| 0.0056 | 0.799 | 2.28 | **0.420** | **0.75** | 0.488 | 0.89 | 0.487 | 0.89 | 0.462 | 0.81 |
| 0.01 | 1.383 | 3.89 | **0.536** | **0.84** | 0.600 | 0.93 | 0.599 | 0.93 | 0.581 | 0.89 |
| 0.018 | 2.435 | 6.97 | **0.651** | **0.90** | 0.700 | 0.94 | 0.700 | 0.94 | 0.689 | 0.93 |
| 0.032 | 4.284 | 12.08 | **0.754** | **0.95** | 0.788 | 0.96 | 0.788 | 0.96 | 0.780 | 0.96 |
| 0.056 | 7.593 | 21.17 | **0.837** | **0.96** | 0.863 | 0.97 | 0.863 | 0.97 | 0.854 | 0.97 |
| 0.1 | 13.470 | 37.47 | **0.894** | **0.99** | 0.911 | **0.99** | 0.911 | **0.99** | 0.904 | **0.99** |
| 0.18 | 23.878 | 67.75 | **0.938** | **1.00** | 0.950 | 1.01 | 0.950 | 1.01 | 0.944 | 1.01 |
| 0.32 | 42.519 | 118.15 | **0.960** | **1.00** | 0.968 | **1.00** | 0.968 | **1.00** | 0.964 | **1.00** |
| 0.56 | 75.568 | 216.32 | **0.979** | **1.00** | 0.984 | 1.01 | 0.984 | 1.01 | 0.981 | 1.01 |
| 1 | 134.283 | 377.26 | **0.985** | **1.01** | 0.988 | **1.01** | 0.988 | **1.01** | 0.987 | **1.01** |
| 0.0001 | **0.054** | **0.17** | **0.054** | **0.17** | 0.085 | 0.27 | 0.086 | 0.27 | 0.102 | 0.31 |
| 0.00018 | 0.065 | **0.20** | **0.064** | **0.20** | 0.094 | 0.29 | 0.095 | 0.29 | 0.109 | 0.32 |
| 0.00032 | 0.084 | 0.26 | **0.081** | **0.24** | 0.111 | 0.33 | 0.112 | 0.33 | 0.124 | 0.35 |
| 0.00056 | 0.118 | 0.36 | **0.109** | **0.31** | 0.140 | 0.40 | 0.141 | 0.40 | 0.149 | 0.41 |
| 0.001 | 0.176 | 0.51 | **0.152** | **0.40** | 0.184 | 0.49 | 0.184 | 0.49 | 0.186 | 0.47 |
| 0.0018 | 0.281 | 0.82 | **0.217** | **0.51** | 0.253 | 0.62 | 0.253 | 0.62 | 0.248 | 0.57 |
| 0.0032 | 0.467 | 1.35 | **0.305** | **0.64** | 0.347 | 0.75 | 0.346 | 0.75 | 0.338 | 0.70 |
| 0.0056 | 0.794 | 2.27 | **0.411** | **0.75** | 0.452 | 0.84 | 0.452 | 0.83 | 0.447 | 0.80 |
| 0.01 | 1.381 | 3.86 | **0.532** | **0.83** | 0.567 | 0.88 | 0.567 | 0.88 | 0.567 | 0.87 |
| 0.018 | 2.430 | 6.86 | **0.649** | **0.89** | 0.677 | 0.91 | 0.677 | 0.91 | 0.676 | 0.91 |
| 0.032 | 4.312 | 12.12 | **0.751** | **0.95** | 0.773 | 0.96 | 0.773 | 0.96 | 0.770 | 0.96 |
| 0.056 | 7.609 | 21.66 | **0.830** | **0.96** | 0.848 | 0.97 | 0.847 | 0.97 | 0.844 | 0.97 |
| 0.1 | 13.479 | 39.01 | **0.894** | **0.98** | 0.907 | **0.98** | 0.907 | **0.98** | 0.903 | **0.98** |
| 0.18 | 23.800 | 66.42 | **0.939** | **0.99** | 0.948 | 1.00 | 0.948 | 1.00 | 0.944 | **0.99** |
| 0.32 | 42.257 | 118.05 | **0.963** | **1.01** | 0.969 | **1.01** | 0.969 | **1.01** | 0.966 | **1.01** |
| 0.56 | 75.882 | 215.48 | **0.976** | **1.00** | 0.979 | **1.00** | 0.979 | **1.00** | 0.978 | **1.00** |
| 1 | 134.283 | 382.26 | **0.990** | **1.02** | 0.992 | **1.02** | 0.992 | **1.02** | 0.991 | **1.02** |

Table S4. Simulation 3: exact values behind Figure S3, right. Mean relative risk against the number of hospitals $K$, for the true $\tau^2$, DerSimonian-Laird [DerSimonian and Laird 1986], and the posterior mean under a half-Cauchy(0.5) prior on $\tau$.

| $K$ | True $\tau^2$ | DerSimonian-Laird [DerSimonian and Laird 1986] | Posterior mean |
|---|---|---|---|
| 3 | **0.777** | 0.911 | 0.928 |
| 4 | **0.772** | 0.923 | 0.884 |
| 6 | **0.736** | 0.869 | 0.806 |
| 8 | **0.715** | 0.817 | 0.770 |
| 12 | **0.669** | 0.766 | 0.719 |
| 16 | **0.653** | 0.733 | 0.701 |
| 24 | **0.657** | 0.701 | 0.691 |
| 48 | **0.658** | 0.678 | 0.672 |

*Simulation 6: structured heterogeneity.* The 24 hospitals are split into four regions of six; $\theta_k$ is the sum of a regional effect with variance $\tau_r^2$ and a hospital effect with variance $\tau_w^2 = 0.002$. The GMRF prior of Equation (5) links hospitals of the same region and is computed by Jacobi iterations on the

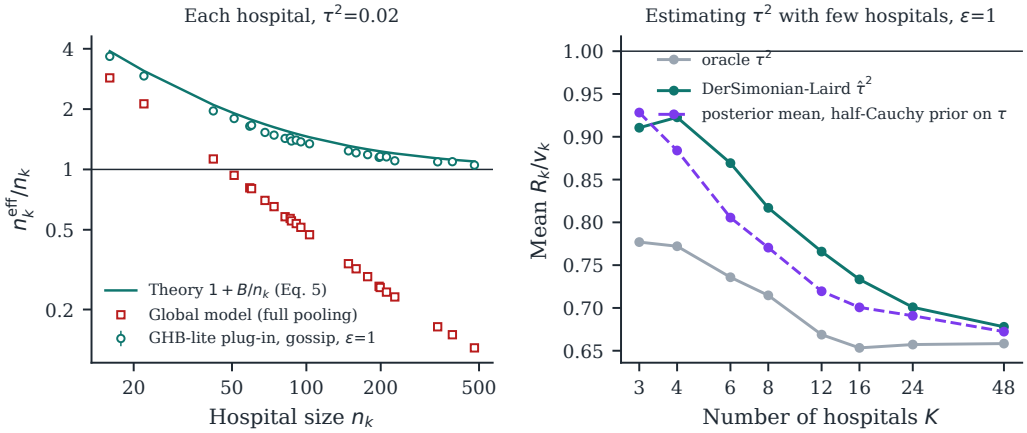



Fig. S3. Simulation 3. Left: effective sample size relative to own size for each hospital, plug-in GHB-lite and a global model; the line is $1 + B/n_k$. Right: mean relative risk against the number of hospitals for three ways of obtaining $\tau^2$.

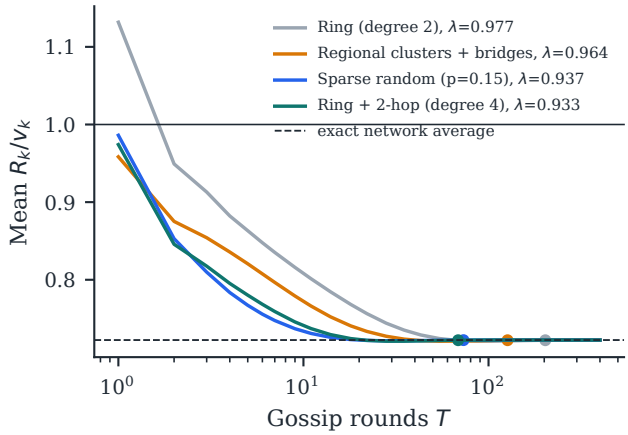



Fig. S4. Simulation 4. Mean relative risk of plug-in GHB-lite against the number of gossip rounds ($\tau^2 = 0.02$, $\epsilon = 1$). Dots mark $\log(100)/(1-\lambda)$; the dashed line is the exact network average.

privatized releases, followed by each hospital's local update with its own noise-free estimate. Its anchor term is switched off so that it uses the same information as the exchangeable prior, and its smoothing weight $\beta$ was chosen on separate tuning replicates (the same value, $\beta \approx 46$, was selected at every $\tau_r^2$); the regions are assumed known. Figure S6 and Table S7 show a crossover. When regions barely differ ($\tau_r^2 \lesssim 0.003$), the exchangeable prior is better (0.31 against 0.40 at the smallest $\tau_r^2$), because the GMRF borrows only from five neighbors. When regional differences dominate, the exchangeable prior loses its gain (0.94) while the GMRF keeps it (0.40).

*Simulation 7: logistic-regression heads.* This simulation leaves the Gaussian model. Each hospital has its own logistic model on four standardized features and an intercept, $\theta_k = \mu + u_k$ with

Table S5. Simulation 4: exact values behind Figure S4. Mean relative risk of plug-in GHB-lite after $T$ gossip rounds on each topology; below, the second-largest eigenvalue modulus $\lambda$ of the gossip matrix, the rounds after which the risk is within 2% of exact averaging, and the rounds that Proposition 4.3 requires for a consensus error of 1%.

| $T$ | Ring | Regional | Sparse random | Ring + 2-hop |
|---|---|---|---|---|
| 1 | 1.132 | **0.958** | 0.986 | 0.974 |
| 2 | 0.949 | 0.875 | 0.853 | **0.846** |
| 3 | 0.913 | 0.854 | **0.810** | 0.818 |
| 4 | 0.882 | 0.836 | **0.784** | 0.795 |
| 5 | 0.863 | 0.821 | **0.767** | 0.780 |
| 6 | 0.848 | 0.808 | **0.756** | 0.768 |
| 7 | 0.835 | 0.797 | **0.748** | 0.759 |
| 9 | 0.816 | 0.779 | **0.737** | 0.746 |
| 11 | 0.801 | 0.766 | **0.731** | 0.737 |
| 14 | 0.785 | 0.752 | **0.726** | 0.729 |
| 18 | 0.768 | 0.741 | **0.723** | 0.724 |
| 22 | 0.756 | 0.733 | **0.722** | **0.722** |
| 28 | 0.744 | 0.727 | **0.721** | **0.721** |
| 35 | 0.735 | 0.724 | **0.721** | **0.721** |
| 43 | 0.729 | 0.722 | 0.722 | **0.721** |
| 54 | 0.724 | **0.721** | 0.722 | 0.722 |
| 68 | 0.722 | **0.721** | 0.722 | 0.722 |
| 85 | **0.721** | **0.721** | 0.722 | 0.722 |
| 106 | **0.721** | 0.722 | 0.722 | 0.722 |
| 132 | **0.721** | 0.722 | 0.722 | 0.722 |
| 165 | **0.722** | **0.722** | **0.722** | **0.722** |
| 206 | **0.722** | **0.722** | **0.722** | **0.722** |
| 257 | **0.722** | **0.722** | **0.722** | **0.722** |
| 320 | **0.722** | **0.722** | **0.722** | **0.722** |
| 400 | **0.722** | **0.722** | **0.722** | **0.722** |
| $\lambda$ | 0.977 | 0.964 | 0.937 | **0.933** |
| rounds to 2% | 35 | 22 | **11** | 14 |
| bound, $\gamma = 0.01$ | 203 | 127 | 73 | **69** |
| exact averaging | | | 0.722 | |

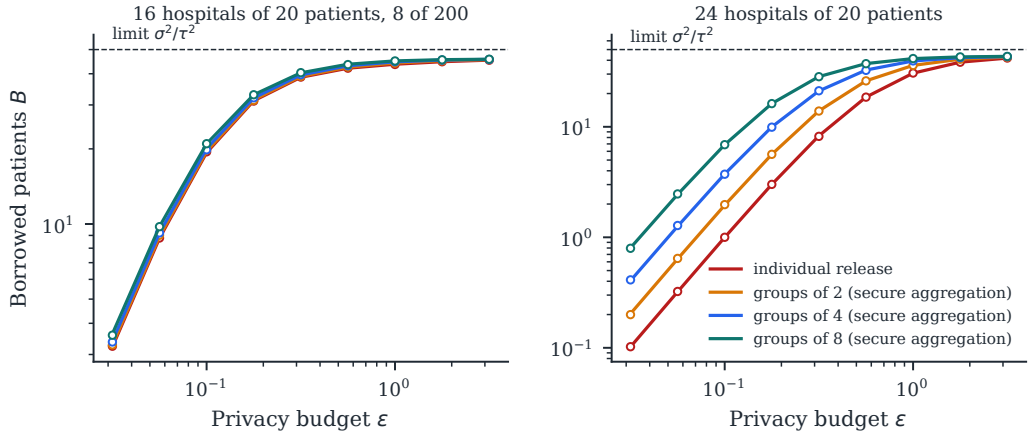



Fig. S5. Simulation 5. Borrowed patients $B$ against the privacy budget for individual and joint release; lines are closed forms, markers simulation.

$u_k \sim \mathcal{N}(0, \tau^2 I)$, and features are clipped so that $\|x\| \le \sqrt{5}$. Here $\mu_0$ is a published model (channel P3), equal to $\mu$ plus noise of standard deviation 0.3. Each hospital releases its mean score at $\mu_0$,

Table S6. Simulation 5: exact values behind Figure S5. Borrowed patients $B$ for groups of $g$ hospitals releasing jointly, simulated, with the closed form in parentheses. All small: 24 hospitals of 20 patients; mixed: 16 of 20 and 8 of 200 ($\tau^2 = 0.02$).

| Network | $\epsilon$ | $g = 1$ | $g = 2$ | $g = 4$ | $g = 8$ |
|---|---|---|---|---|---|
| all small | 0.0316 | 0.1 (0.1) | 0.2 (0.2) | 0.4 (0.4) | 0.8 (0.8) |
| | 0.0562 | 0.3 (0.3) | 0.6 (0.6) | 1.3 (1.3) | 2.5 (2.4) |
| | 0.1 | 1.0 (1.0) | 2.0 (2.0) | 3.7 (3.7) | 6.9 (6.9) |
| | 0.178 | 3.0 (3.0) | 5.6 (5.6) | 9.9 (10.0) | 16.2 (16.2) |
| | 0.316 | 8.2 (8.3) | 13.9 (13.9) | 21.2 (21.1) | 28.4 (28.5) |
| | 0.562 | 18.6 (18.6) | 26.1 (26.1) | 32.4 (32.6) | 37.3 (37.3) |
| | 1 | 30.7 (30.6) | 35.9 (36.0) | 39.5 (39.4) | 41.5 (41.4) |
| | 1.78 | 38.4 (38.4) | 40.9 (40.9) | 42.3 (42.2) | 42.9 (42.9) |
| | 3.16 | 41.9 (41.9) | 42.6 (42.7) | 43.1 (43.2) | 43.3 (43.4) |
| mixed | 0.0316 | 3.2 (3.2) | 3.3 (3.3) | 3.4 (3.4) | 3.6 (3.6) |
| | 0.0562 | 8.8 (8.8) | 9.0 (8.9) | 9.2 (9.2) | 9.8 (9.8) |
| | 0.1 | 19.5 (19.3) | 19.8 (19.6) | 19.8 (20.1) | 21.0 (21.0) |
| | 0.178 | 31.1 (31.2) | 31.3 (31.5) | 32.0 (32.0) | 33.0 (33.0) |
| | 0.316 | 38.8 (38.7) | 39.1 (39.0) | 39.6 (39.6) | 40.4 (40.4) |
| | 0.562 | 42.1 (42.1) | 42.6 (42.5) | 43.1 (43.0) | 43.5 (43.7) |
| | 1 | 43.6 (43.7) | 44.1 (44.2) | 44.7 (44.6) | 45.0 (45.0) |
| | 1.78 | 44.7 (44.7) | 45.1 (45.1) | 45.4 (45.4) | 45.5 (45.6) |
| | 3.16 | 45.3 (45.4) | 45.6 (45.6) | 45.6 (45.7) | 45.7 (45.7) |

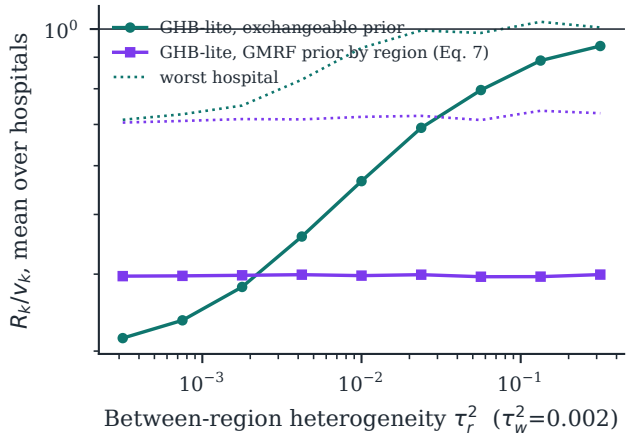



Fig. S6. Simulation 6. Mean relative risk (solid) and worst hospital (dotted) for the exchangeable and the regional GMRF prior, against the between-region variance ($\epsilon = 1$).

$g_k = \frac{1}{n_k}\sum_i x_i(y_i - \sigma(x_i^\top \mu_0))$, plus isotropic Gaussian noise. Its $\ell_2$ sensitivity is $2\sqrt{5}/n_k$, and the noise is calibrated with zero-concentrated DP: $s_k = \Delta_k/\sqrt{2\rho}$ with $\rho + 2\sqrt{\rho \ln(1/\delta)} = \epsilon$. Under the public feature distribution, $\mathbb{E}g_k = m(\theta_k) - m(\mu_0)$ with $m(\theta) = \mathbb{E}[x\,\sigma(x^\top\theta)]$, so the releases are mapped to the parameter scale by moment matching, linearized first at the pooled moment estimate and then at each hospital's leave-one-out network estimate; their variances follow by the delta method. The anchor shapes that second point only for hospitals where it passes the conflict check against the anchor-free network estimate. GHB-lite then borrows coordinatewise. Each hospital summarizes the posterior of $\tau$ (half-Cauchy(0.5) prior) by 12 quantiles, combines each component's network estimate with $\mu_0$ by precision weighting unless it dropped the anchor, multiplies by the weakly informative default $\mathcal{N}(0, 2.5^2)$ [Gelman et al. 2008], and averages its maximum a posteriori

Table S7. Simulation 6: exact values behind Figure S6. Mean and worst relative risk for the exchangeable prior and the regional GMRF prior against the between-region variance $\tau_r^2$ ($\epsilon = 1$).

| | Mean | | Worst hospital | |
|---|---|---|---|---|
| $\tau_r^2$ | Exchangeable | GMRF | Exchangeable | GMRF |
| 0.00032 | **0.315** | 0.397 | 0.71 | **0.70** |
| 0.00075 | **0.337** | 0.397 | 0.73 | **0.71** |
| 0.0018 | **0.381** | 0.398 | 0.75 | **0.71** |
| 0.0042 | 0.460 | **0.399** | 0.83 | **0.71** |
| 0.01 | 0.566 | **0.398** | 0.93 | **0.72** |
| 0.024 | 0.691 | **0.399** | 0.99 | **0.72** |
| 0.056 | 0.796 | **0.396** | 0.99 | **0.71** |
| 0.13 | 0.889 | **0.396** | 1.03 | **0.74** |
| 0.32 | 0.939 | **0.399** | 1.01 | **0.73** |

fits under the components, weighted by the Laplace approximation of the marginal likelihood of its own data (Remark 4.4). The code is `simulations/ghb_logit.py`. The baselines are a local model with prior $\mathcal{N}(\mu_0, I)$, which uses the anchor but no other hospital, and a global model fitted on the pooled data of all hospitals without privacy. Table S8 and Figure S7 report the excess log-loss over the true model on 4,000 new patients per hospital, averaged over 100 replicates. At $\tau^2 = 0.05$ and $\epsilon = 1$, GHB-lite reduces the excess log-loss of local estimation by 49% (0.0266 to 0.0135), below the global model trained without privacy (0.0173), and no hospital is worse off on average (largest ratio to local 0.85, against 2.48 for the global model). Without privacy GHB-lite reaches 0.0106, better than the global model. At $\epsilon = 0.3$ it keeps most of its gain (0.0175 against 0.0273).

*Development note (first version).* Three components of an earlier version of this simulation were added after the first runs failed. The first version used the ridge releases without undoing the shrinkage, estimated $\tau^2$ by DerSimonian-Laird [DerSimonian and Laird 1986] and ignored $\mu_0$. With privacy noise at $\epsilon = 1$ its estimates of $\tau^2$ were erratic (median 0, mean several times the truth), and it was worse than local estimation at $\tau^2 = 0.2$ and at $\epsilon = 0.3$. Each component follows the design of the paper (post-processing, Remark 4.4 and layer L3), but they were chosen with knowledge of these failures, and the results should be read accordingly. A fourth change came from Simulation 11: the releases were first centred on the anchor ($c = \mu_0$), and the shrinkage was undone with the Fisher information at $\mu_0$. When the anchor was badly wrong, this corrupted every release, and GHB-lite without the anchor reached an excess log-loss of 0.34 against 0.06 for local estimation at an anchor error of two standard deviations. Centring on a fixed reference removes the dependence; Simulations 7, 8, 10 and 11 are reported with it. With anchor-centred releases, Simulation 7 had given a slightly smaller gain at $\epsilon = 1$ and no gain at $\epsilon = 0.3$. The first version of Simulation 6 also gave the GMRF prior the anchor $\mu_0$, which the exchangeable prior did not have; the version reported here removes it.

*Simulation 8: federated and meta-analytic baselines.* The logistic world of Simulation 7, at $\epsilon = 1$ and four levels of heterogeneity. All private methods use the same releases, so they satisfy the same guarantee: one-shot federated averaging [McMahan et al. 2017] (the sample-size weighted mean of the releases, one model for every hospital), the random-effects meta-analytic mean (DerSimonian-Laird [DerSimonian and Laird 1986] weights, one model for every hospital), and a Ditto-style [Li et al. 2021] personalized model, which fits each hospital's data with a proximal term $\frac{\lambda}{2}\|\theta - \bar{\theta}_{\text{FedAvg}}\|^2$. The weight $\lambda$ is chosen from $\{0.3, 1, 3, 10, 30, 100\}$ on ten separate tuning replicates per setting, which favours Ditto-style [Li et al. 2021]. The global model fitted on the pooled data without privacy

Table S8. Simulation 7: exact values behind Figure S7. Excess log-loss over the true model ($10^{-3}$ nats) and AUC, mean over 24 hospitals and 100 replicates, and the largest ratio of any hospital's mean excess log-loss to its local one (Worst). Oracle: the true model of each hospital. Global: one model on the pooled data, without privacy.

| | | Excess log-loss | | | AUC | | | | Worst | |
|---|---|---|---|---|---|---|---|---|---|---|
| $\tau^2$ | $\epsilon$ | Local | Global | GHB-lite | Local | Global | GHB-lite | Oracle | GHB-lite | Global |
| 0.01 | 1.0 | 28.3 | **4.1** | 10.1 | 0.715 | **0.733** | 0.728 | 0.737 | 0.82 | **0.77** |
| 0.05 | 1.0 | 26.6 | 17.3 | **13.5** | 0.723 | 0.728 | **0.731** | 0.743 | **0.85** | 2.48 |
| 0.2 | 1.0 | 26.8 | 60.1 | **21.4** | 0.749 | 0.717 | **0.751** | 0.770 | **1.02** | 9.26 |
| 0.5 | 1.0 | **27.1** | 125.7 | 28.5 | **0.784** | 0.688 | 0.781 | 0.804 | **1.17** | 20.68 |
| 0.01 | $\infty$ | 27.4 | **4.2** | **4.2** | 0.717 | **0.734** | 0.733 | 0.737 | **0.53** | 0.68 |
| 0.05 | $\infty$ | 28.1 | 17.0 | **10.6** | 0.722 | 0.729 | **0.734** | 0.743 | **0.85** | 2.71 |
| 0.2 | $\infty$ | 27.4 | 59.9 | **19.9** | 0.745 | 0.712 | **0.748** | 0.766 | **1.02** | 9.56 |
| 0.5 | $\infty$ | 27.7 | 125.9 | **26.0** | **0.785** | 0.690 | **0.785** | 0.805 | **1.07** | 19.34 |
| 0.05 | 0.3 | 27.3 | **16.7** | 17.5 | 0.725 | **0.730** | **0.730** | 0.745 | **0.90** | 2.50 |
| 0.05 | 3.0 | 27.1 | 16.9 | **11.3** | 0.722 | 0.728 | **0.733** | 0.743 | **0.87** | 2.69 |
| 0.05 | 10.0 | 28.2 | 17.0 | **10.9** | 0.723 | 0.729 | **0.734** | 0.743 | **0.84** | 2.39 |

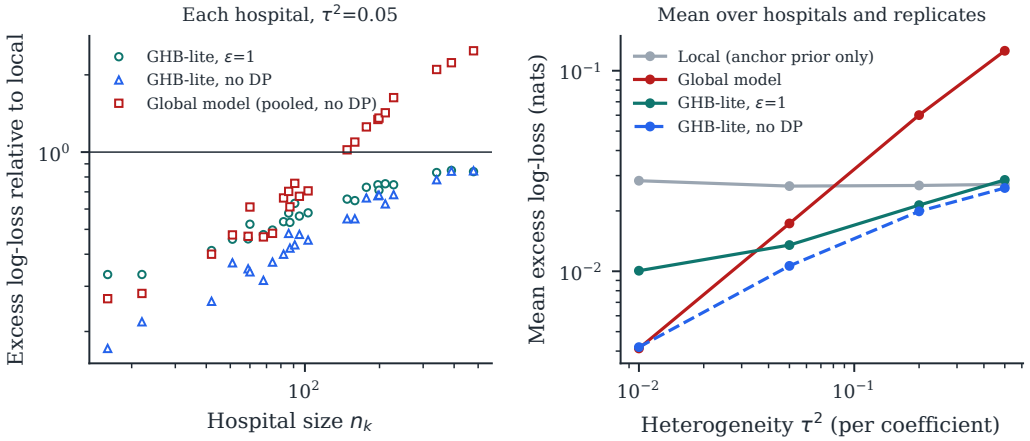



Fig. S7. Simulation 7. Left: excess log-loss relative to local estimation for each hospital ($\tau^2 = 0.05$). Right: mean excess log-loss against heterogeneity.

is a reference that no private method can match in general. Results (Figure S8, left, and Table 3 of the paper; 40 replicates per setting): at $\tau^2 = 0.05$, GHB-lite 0.0134, Ditto-style [Li et al. 2021] 0.0156, pooled 0.0167, meta-analysis 0.0303, federated averaging [McMahan et al. 2017] 0.0365, local 0.0274. At $\tau^2 = 0.5$, GHB-lite 0.0282, Ditto-style [Li et al. 2021] 0.0265, local 0.0273. One-shot federated averaging [McMahan et al. 2017] is weak here because it averages noisy releases without personalization; multi-round federated training with differentially private gradients is compared in Simulation 15.

*Simulation 10: case-mix shift.* As Simulation 8 at $\tau^2 = 0.05$, but each hospital draws its features from $\mathcal{N}(m_k, I)$ with $m_k \sim \mathcal{N}(0, s^2 I)$, while the de-shrinking still uses the public, unshifted feature distribution (Figure S8, right, and Table 3 of the paper). At $s = 1$: GHB-lite 0.0132, Ditto-style [Li et al. 2021] 0.0140, pooled 0.0198, local 0.0228.

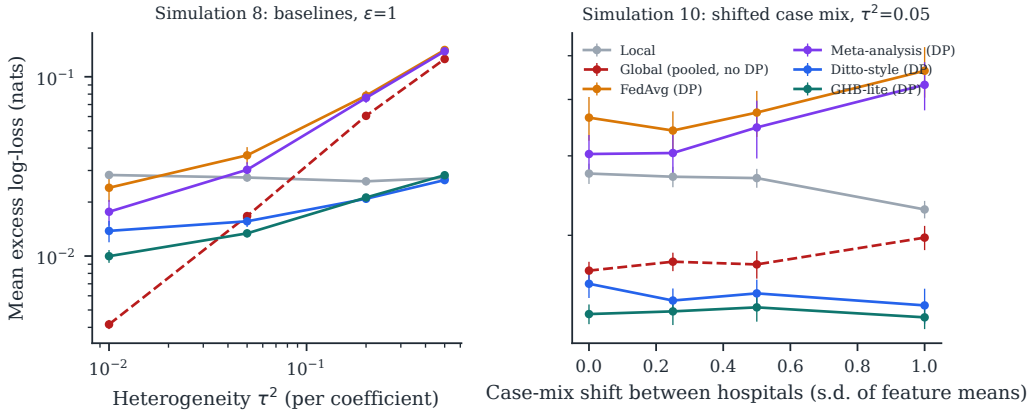



Fig. S8. Simulations 8 and 10. Mean excess log-loss of GHB-lite and of private and non-private baselines against heterogeneity (left) and case-mix shift (right). Bars: ±2 standard errors.

Table S9. Simulation 9: exact values behind Figure S9, right. Mean relative risk; with an outlier, for the outlying hospital and for the others.

| | All hospitals, or the outlier | | | The others | | |
|---|---|---|---|---|---|---|
| Site effects | GHB-lite | Robust | Opt-out | GHB-lite | Robust | Opt-out |
| Gaussian effects | **0.722** | 0.763 | 0.728 | | | |
| Student-$t_3$ effects | 0.685 | **0.653** | 0.667 | | | |
| outlier at the smallest hospital | 6.66 | 4.60 | **4.36** | **0.731** | 0.771 | 0.738 |
| outlier at the largest hospital | 1.11 | **1.06** | 1.11 | 0.850 | **0.748** | 0.850 |

*Simulation 9: atypical hospitals.* The network of Simulation 2 at $\tau^2 = 0.02$ and $\epsilon = 1$, with Gaussian site effects, Student-$t_3$ site effects of the same variance, or Gaussian effects plus one hospital at $u = 6\tau$, placed at the smallest or at the largest hospital. Three estimators: Gaussian GHB-lite; robust GHB-lite, which replaces the Gaussian prior $\mathcal{N}(\tilde{\mu}_k, \hat{\tau}^2 + M_k)$ by a Student-$t_3$ prior of the same variance (posterior mean by quadrature); and GHB-lite with opt-out, where a hospital keeps its local estimate when $|\hat{\theta}_k - \tilde{\mu}_k| / \sqrt{v_k + \hat{\tau}^2 + M_k} > 3$. Both variants change only the hospital's own computation and cost no privacy. Mean relative risk: Gaussian effects 0.72 (robust 0.76), $t_3$ effects 0.69 (robust 0.65). The outlying hospital: 6.7 (robust 4.6, opt-out 4.4) when it is the smallest, 1.11 (robust 1.06) when it is the largest (Figure S9; Tables S9 and S10). With 16 patients, the standard error of the local estimate is as large as the deviation, so no test based on the hospital's own data can detect it reliably.

*Simulation 11: a wrong anchor.* The logistic world at $\tau^2 = 0.05$ and $\epsilon = 1$, with the anchor wrong by $\mathcal{N}(0, e^2 I)$ for $e$ from 0 to 3. GHB-lite uses the anchor with its uncertainty stated honestly ($e$), overconfidently (0.3 whatever $e$), overconfidently with a conflict check (the anchor is dropped when $|\mu_0 - \tilde{\mu}_k| > 3\sqrt{0.3^2 + M_k}$ in any coordinate), or not at all. The local model uses $\mathcal{N}(\mu_0, I)$ as its prior and degrades with the anchor (Figure S10 and Table S11). Honest uncertainty and the conflict check keep the excess log-loss at or below 0.0166 at every error, GHB-lite without the anchor at or below 0.0167; an overconfident anchor two standard deviations off reaches 0.0352, and local estimation 0.0605.

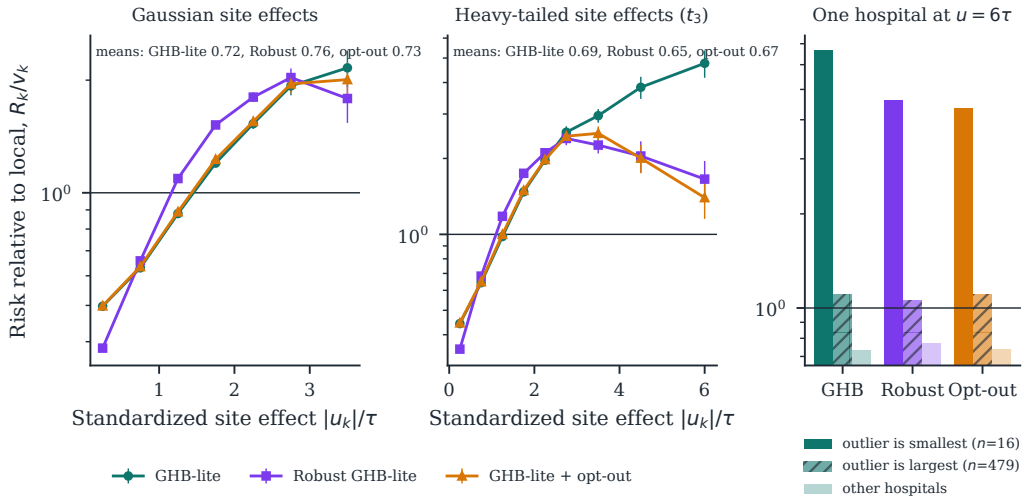



Fig. S9. Simulation 9. Left and centre: relative risk against the standardized site effect for Gaussian and heavy-tailed effects. Right: the outlying hospital and the others, when the outlier is the smallest or the largest hospital.

Table S10. Simulation 9: exact values behind Figure S9, centre. Relative risk with Student-$t_3$ site effects, by standardized effect $|u_k|/\tau$, with two standard errors in grey; $N$: hospital-replicates in the bin.

| $\lvert u_k\rvert/\tau$ | GHB-lite | Robust | Opt-out | $N$ |
|---|---|---|---|---|
| 0-0.5 | 0.44 (0.01) | **0.35** (0.00) | 0.45 (0.01) | 79,111 |
| 0.5-1 | **0.64** (0.01) | 0.68 (0.01) | 0.65 (0.01) | 38,820 |
| 1-1.5 | **0.98** (0.02) | 1.18 (0.02) | 1.00 (0.02) | 14,436 |
| 1.5-2 | **1.47** (0.04) | 1.75 (0.05) | 1.49 (0.04) | 5,717 |
| 2-2.5 | **1.97** (0.08) | 2.11 (0.09) | 1.98 (0.08) | 2,604 |
| 2.5-3 | 2.54 (0.14) | **2.40** (0.14) | 2.45 (0.14) | 1,263 |
| 3-4 | 2.96 (0.18) | **2.26** (0.17) | 2.52 (0.16) | 1,109 |
| 4-5 | 3.83 (0.38) | 2.04 (0.29) | **2.01** (0.26) | 467 |
| 5-7 | 4.77 (0.59) | 1.66 (0.30) | **1.40** (0.25) | 322 |

Table S11. Simulation 11: exact values behind Figure S10. Mean excess log-loss ($10^{-3}$ nats) against the standard deviation $e$ of the anchor's error, with two standard errors in grey; Honest: stated uncertainty $e$; Over.: stated 0.3; + check: stated 0.3 with the conflict check; Dropped: percentage of hospitals where the check drops the anchor.

| $e$ | Local | Honest | Over. | Over. + check | No anchor | Dropped (%) |
|---|---|---|---|---|---|---|
| 0.0 | 27.0 (1.5) | **11.4** (0.6) | 13.5 (1.0) | 13.5 (1.0) | 16.1 (2.1) | 0 |
| 0.3 | 27.4 (1.7) | 14.1 (0.9) | 14.1 (0.9) | **14.0** (0.9) | 14.9 (1.0) | 2 |
| 1.0 | 31.6 (1.8) | **14.5** (0.9) | 19.0 (2.0) | 15.5 (1.0) | 14.9 (0.9) | 74 |
| 2.0 | 60.5 (9.1) | **16.6** (2.2) | 35.2 (3.7) | 16.7 (2.2) | 16.7 (2.2) | 100 |
| 3.0 | 96.9 (12.6) | **14.8** (1.1) | 40.5 (3.7) | 14.9 (1.1) | 14.9 (1.1) | 100 |

*Simulation 12: dishonest hospitals.* The network of Simulation 2 at $\tau^2 = 0.02$ and $\epsilon = 1$. The one or three largest hospitals release $\mu + b\tau$ instead of their privatized estimate; sizes are public, so they cannot change their weights. Robust GHB-lite obtains all releases by flooding, computes an inverse-variance weighted median and a MAD-based spread, drops releases more than three spreads

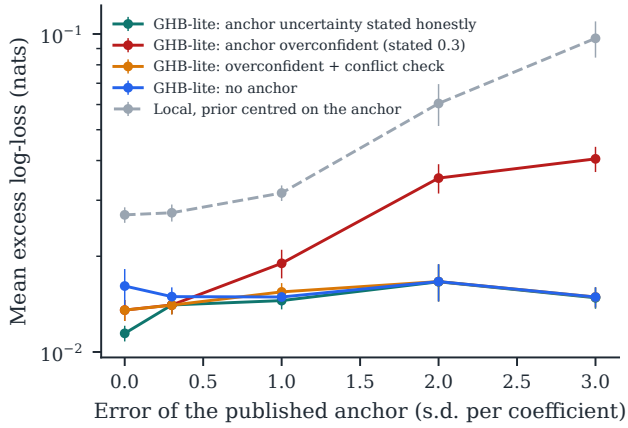



Fig. S10. Simulation 11. Mean excess log-loss against the error of the published anchor.

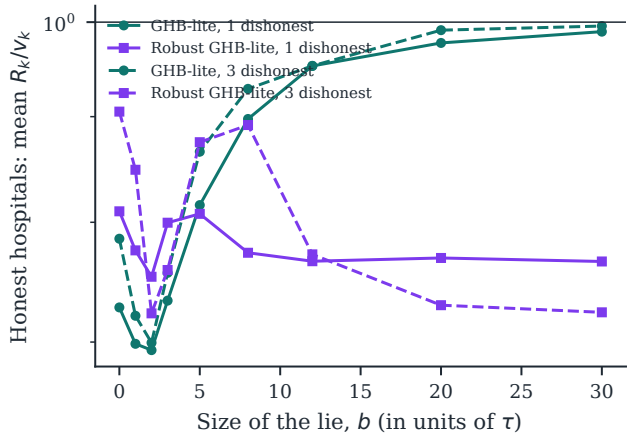



Fig. S11. Simulation 12. Honest hospitals' mean relative risk against the size of the lie, for one and three dishonest hospitals.

away, and runs GHB-lite on the rest; this is post-processing. Honest hospitals' mean relative risk (Figure S11 and Table S12): plain GHB-lite never exceeds 1.00, because a large lie inflates $\hat{\tau}^2$ and switches borrowing off; robust GHB-lite keeps 0.72 against large lies, but costs 0.81 against 0.73 when nobody lies, because the screen sometimes drops honest outlying releases.

*Simulation 13: link failures.* The network of Simulation 2 with every link failing independently in each round with probability $p$; the Metropolis-Hastings weights are recomputed on the working links, so each round's matrix stays symmetric and doubly stochastic. With $p = 0.8$, the mean relative risk is 0.77 after 20 rounds and 0.73 after 60, against 0.72 with exact averaging (Figure S12 and Table S13).

*Simulation 14: repeated releases.* For a total budget of $\epsilon = 1$, $\delta = 10^{-5}$ spread over $r$ refits on the same patients, the borrowed patients $B$ after the last refit (closed form, network of Simulation

Table S12. Simulation 12: exact values behind Figure S11. Honest hospitals' mean relative risk against the size $b$ of the lie (in units of $\tau$), and the percentage of dishonest releases that robust GHB-lite keeps.

| | One dishonest hospital | | | Three dishonest hospitals | | |
|---|---|---|---|---|---|---|
| $b$ | GHB-lite | Robust | Kept (%) | GHB-lite | Robust | Kept (%) |
| 0 | **0.728** | 0.810 | 100 | **0.785** | 0.905 | 100 |
| 1 | **0.699** | 0.775 | 99 | **0.721** | 0.848 | 100 |
| 2 | **0.694** | 0.753 | 88 | **0.699** | 0.723 | 100 |
| 3 | **0.733** | 0.800 | 73 | **0.756** | 0.759 | 99 |
| 5 | 0.815 | **0.808** | 28 | **0.866** | 0.875 | 95 |
| 8 | 0.898 | **0.773** | 2 | 0.928 | **0.892** | 64 |
| 12 | 0.952 | **0.766** | 0 | 0.952 | **0.772** | 18 |
| 20 | 0.977 | **0.769** | 0 | 0.991 | **0.729** | 0 |
| 30 | 0.989 | **0.766** | 0 | 0.996 | **0.724** | 0 |

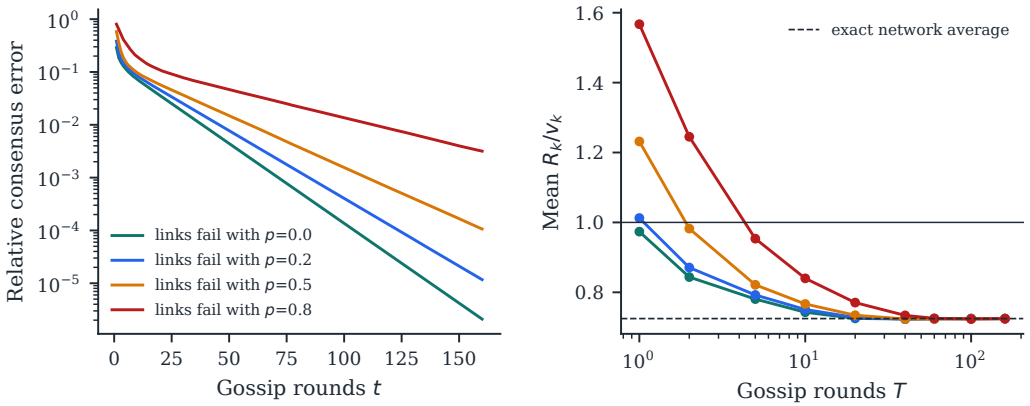



Fig. S12. Simulation 13. Left: consensus error against rounds. Right: mean relative risk of plug-in GHB-lite against rounds, for four link-failure probabilities.

Table S13. Simulation 13: exact values behind Figure S12, right. Mean relative risk of plug-in GHB-lite after $T$ rounds, for link-failure probability $p$.

| $T$ | $p = 0$ | $p = 0.2$ | $p = 0.5$ | $p = 0.8$ |
|---|---|---|---|---|
| 1 | 0.974 | 1.013 | 1.232 | 1.567 |
| 2 | 0.844 | 0.871 | 0.982 | 1.245 |
| 5 | 0.781 | 0.792 | 0.821 | 0.954 |
| 10 | 0.743 | 0.751 | 0.767 | 0.840 |
| 20 | 0.725 | 0.728 | 0.734 | 0.771 |
| 40 | 0.724 | 0.723 | 0.724 | 0.734 |
| 60 | 0.724 | 0.724 | 0.724 | 0.726 |
| 100 | 0.725 | 0.725 | 0.725 | 0.723 |
| 160 | 0.725 | 0.725 | 0.725 | 0.724 |
| exact | | 0.725 | | |

2) under basic composition ($(\epsilon/r, \delta/r)$ per release) and under zero-concentrated accounting (the Gaussian mechanism with noise $s$ is $\rho$-zCDP with $\rho = \Delta^2/(2s^2)$; costs add; $\rho$-zCDP implies ($\rho$ +

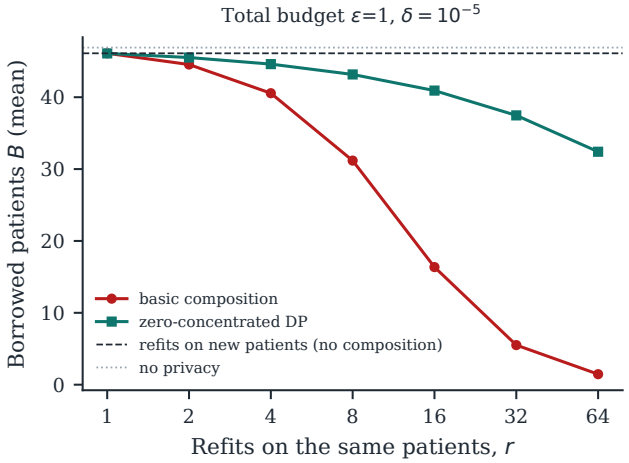



Fig. S13. Simulation 14. Borrowed patients after $r$ refits on the same patients, for a fixed total budget.

Table S14. Simulation 14: exact values behind Figure S13. Borrowed patients $B$ after $r$ refits for a total budget $\epsilon = 1$, $\delta = 10^{-5}$, under basic composition and zero-concentrated accounting, and the ratio of the per-release noise standard deviations (basic over zCDP).

| $r$ | $B$, basic | $B$, zCDP | Noise ratio |
|---|---|---|---|
| 1 | **46.1** | **46.1** | 0.99 |
| 2 | 44.5 | **45.5** | 1.44 |
| 4 | 40.6 | **44.6** | 2.09 |
| 8 | 31.2 | **43.2** | 3.03 |
| 16 | 16.4 | **40.9** | 4.40 |
| 32 | 5.5 | **37.5** | 6.36 |
| 64 | 1.5 | **32.4** | 9.20 |
| no privacy | | $B$ = 46.9 | |

$2\sqrt{\rho \ln(1/\delta)}, \delta)$-DP). After 64 refits, $B = 1.5$ under basic composition and 32.4 under zCDP, against 46.1 for a single release (Figure S13 and Table S14).

*Simulation 15: iterative federated learning.* The logistic world of Simulation 8 (24 hospitals, ring with two-hop links, $\delta = 10^{-5}$), GHB-lite against multi-round training at the same privacy budget per hospital. GHB-lite is the revised version of Section S6; the earlier version is reported alongside it. Per-patient gradients of the logistic loss are $(p - y)x$ with $\|x\| \le \sqrt{5}$, so they need no clipping, and replacing one patient moves a hospital's mean gradient by at most $2\sqrt{5}/n_k$. Every iterative method adds Gaussian noise to each hospital's mean gradient in every round and is accounted with zero-concentrated DP: with $R$ rounds, each round spends $\rho/R$, and the total $\rho$ satisfies $\rho + 2\sqrt{\rho \ln(1/\delta)} = \epsilon$. GHB-lite's single release is accounted in the same way. All methods start from, or shrink toward, the published anchor $\mu_0$, which GHB-lite also uses.

- *DP-FedAvg* [McMahan et al. 2017, 2018], full-batch: a server holds one model; in each round every hospital sends its noisy mean gradient at that model, and the server takes a step of size $\eta$ along the sample-size weighted average.

- *DP-FedAvg [McMahan et al. 2018] with fine-tuning*: each hospital then takes $E$ plain gradient steps (step size 1) on its own data, starting from the global model; this costs no privacy because the data stay local.
- *Decentralized DP-SGD* [Lian et al. 2017], the algorithm behind swarm learning [Warnat-Herresthal et al. 2021]: no server; each hospital steps along its own noisy gradient and averages models with its neighbours using the Metropolis-Hastings weights of GHB-lite's graph (adapt, then combine), followed by optional fine-tuning.
- *Clustered DP-FedAvg*, in the style of IFCA [Ghosh et al. 2020]. IFCA [Ghosh et al. 2020] lets each hospital choose the cluster whose model fits its data best in every round. Under DP this choice has sensitivity of order 1, not $1/n_k$, because one patient can move a hospital's whole gradient from one cluster to another. Hospitals therefore make one private ridge release with half the budget, the releases are clustered by weighted $k$-means (post-processing), and DP-FedAvg [McMahan et al. 2018] runs within each cluster with the other half, followed by optional fine-tuning.
- *Local gradient descent from the anchor*, a control that uses no other hospital: $E$ gradient steps on the hospital's own data from $\mu_0$, with early stopping as the only regularizer.
- *Ditto-style [Li et al. 2021]* as in Simulation 8, and *GHB-lite with the true* $\tau^2$ (the marginal heterogeneity, regional plus within-region in the regional world), a reference that no method can use.

Hyperparameters are chosen on 20 separate replicates per setting by the mean excess log-loss, which requires access to the true models: rounds $R \in \{1, 2, 3, 5, 10, 30, 100\}$ ($\{3, 10, 30, 100\}$ for decentralized training), step size $\eta \in \{0.2, 0.5, 2\}$, fine-tuning steps $E \in \{0, 1, 2, 3, 5, 10, 30, 100\}$ and clusters $C \in \{2, 3, 4\}$. Table S16 lists the choices. The only choices at the edge of a grid are $R = 1$, the smallest meaningful number of rounds, and $\eta = 0.2$ for decentralized training, where smaller steps would approach the anchor-only control. GHB-lite has no tuning parameter. Settings: $\tau^2 \in \{0.01, 0.05, 0.2, 0.5\}$ at $\epsilon = 1$; $\epsilon \in \{0.3, \infty\}$ at $\tau^2 = 0.05$; a regional world in which the 24 hospitals form four regions, with between-region variance 0.2 and within-region variance 0.02 per coefficient, where clustering should help; an anchor wrong by $\mathcal{N}(0, I)$ instead of $\mathcal{N}(0, 0.3^2 I)$; $\tau^2 = 0.1$ at $\epsilon = 0.5$; and a network of 48 hospitals (sizes drawn as for the 24, ring with two-hop links). The last four settings (marked $*$) were not used while revising GHB-lite; the development runs used other seeds, an anchor error of two standard deviations and a three-region world. Results are means over 60 further replicates (Table S15, Figure S14); Table 5 of the paper gives the worst hospital, the share of hospitals that gain, and the largest hospitals.

GHB-lite is the most accurate private method in six of the ten settings ($\tau^2 = 0.01$ and 0.05, $\epsilon = 0.3$, and the held-out regional, $\tau^2 = 0.1/\epsilon = 0.5$ and 48-hospital settings). The tuned Ditto-style [Li et al. 2021] model, which shrinks toward the average of GHB-lite's own releases, is slightly more accurate at $\tau^2 = 0.2$, without privacy and with a wrong anchor; at $\tau^2 = 0.5$ no private method beats local estimation. GHB-lite is more accurate than DP-FedAvg [McMahan et al. 2018] with fine-tuning, decentralized DP-SGD [Lian et al. 2017] and clustered training in every private setting. At $\epsilon = 0.3$, the local control that starts from the anchor reaches 0.0205 against 0.0175 for GHB-lite, and with a wrong anchor it becomes worse than local estimation (0.0419 against 0.0346). GHB-lite's worst hospital is at most 1.06 times its local error in every private setting except $\tau^2 = 0.5$ (1.15), whereas fine-tuning's worst hospital is 1.39 to 2.96 times worse than alone; the hospitals that lose are the large ones. GHB-lite with the true $\tau^2$ remains better still (0.0158 at $\epsilon = 0.3$), so estimating heterogeneity from noisy releases is what is left to improve. The earlier version of GHB-lite reached 0.0259 at $\epsilon = 0.3$ and 0.0166 at $\tau^2 = 0.05$.

*Simulation 16: ablation.* The logistic world of Simulation 15 (24 hospitals, ring with two-hop links, $T = 60$, $\delta = 10^{-5}$), in eight settings: $\tau^2 \in \{0.01, 0.05, 0.2, 0.5\}$ at $\epsilon = 1$; $\epsilon \in \{0.3, \infty\}$ at

Table S15. Simulation 15. Mean excess log-loss (nats) over the hospitals and 60 replicates; lowest of the methods available to a hospital in bold (the earlier version of GHB-lite, the oracle and the pooled model are references). FT: fine-tuning. ∗: setting not used while revising GHB-lite.

| Setting | Local | Anchor GD | GHB-lite | Ditto-style [Li et al. 2021] | DP-FedAvg [McMahan et al. 2018] | DP-FedAvg + FT [McMahan et al. 2018] | Decentr. DP-SGD [Lian et al. 2017] | Clustered DP-FedAvg [Ghosh et al. 2020] | GHB-lite, earlier | GHB-lite, true $\tau^2$ | Pooled, no DP |
|---|---|---|---|---|---|---|---|---|---|---|---|
| $\tau^2 = 0.01$ | 0.0284 | 0.0166 | **0.0101** | 0.0122 | 0.0156 | 0.0114 | 0.0166 | 0.0163 | 0.0155 | 0.0081 | 0.0043 |
| $\tau^2 = 0.05$ | 0.0271 | 0.0190 | **0.0138** | 0.0154 | 0.0274 | 0.0153 | 0.0191 | 0.0214 | 0.0166 | 0.0124 | 0.0173 |
| $\tau^2 = 0.2$ | 0.0267 | 0.0244 | 0.0217 | **0.0205** | 0.0732 | 0.0232 | 0.0244 | 0.0244 | 0.0207 | 0.0194 | 0.0610 |
| $\tau^2 = 0.5$ | **0.0260** | 0.0293 | 0.0275 | 0.0260 | 0.1366 | 0.0287 | 0.0294 | 0.0294 | 0.0273 | 0.0240 | 0.1244 |
| $\epsilon = 0.3$ | 0.0269 | 0.0205 | **0.0175** | 0.0306 | 0.0452 | 0.0196 | 0.0248 | 0.0206 | 0.0259 | 0.0158 | 0.0170 |
| no DP | 0.0278 | 0.0190 | 0.0104 | **0.0101** | 0.0167 | 0.0113 | 0.0118 | 0.0138 | 0.0130 | 0.0097 | 0.0167 |
| regional* | 0.0287 | 0.0265 | **0.0209** | 0.0212 | 0.0631 | 0.0249 | 0.0264 | 0.0271 | 0.0219 | 0.0199 | 0.0485 |
| wrong anchor* | 0.0346 | 0.0419 | 0.0162 | **0.0148** | 0.0446 | 0.0210 | 0.0418 | 0.0463 | 0.0194 | 0.0150 | 0.0169 |
| $\tau^2 = 0.1$, $\epsilon = 0.5$* | 0.0269 | 0.0228 | **0.0184** | 0.0238 | 0.0526 | 0.0217 | 0.0239 | 0.0229 | 0.0227 | 0.0171 | 0.0327 |
| $K = 48$* | 0.0324 | 0.0224 | **0.0144** | 0.0146 | 0.0258 | 0.0160 | 0.0225 | 0.0241 | 0.0170 | 0.0132 | 0.0172 |

Table S16. Simulation 15. Hyperparameters chosen on the tuning replicates: rounds $R$, step size $\eta$, fine-tuning steps $E$, clusters $C$, and the proximal weight $\lambda$ of the Ditto-style [Li et al. 2021] model.

| Setting | DP-FedAvg [McMahan et al. 2018] + FT | Decentralized DP-SGD [Lian et al. 2017] | Clustered DP-FedAvg [Ghosh et al. 2020] | Anchor GD | Ditto [Li et al. 2021] $\lambda$ |
|---|---|---|---|---|---|
| $\tau^2 = 0.01$ | R=3, $\eta$=2.0, E=2 | R=3, $\eta$=0.2, E=5 | C=2, R=1, $\eta$=0.5, E=5 | E=5 | 10 |
| $\tau^2 = 0.05$ | R=10, $\eta$=0.5, E=5 | R=3, $\eta$=0.2, E=5 | C=2, R=5, $\eta$=0.2, E=5 | E=5 | 10 |
| $\tau^2 = 0.2$ | R=2, $\eta$=2.0, E=10 | R=3, $\eta$=0.2, E=10 | C=3, R=1, $\eta$=0.5, E=10 | E=10 | 3 |
| $\tau^2 = 0.5$ | R=30, $\eta$=0.2, E=10 | R=3, $\eta$=0.2, E=10 | C=2, R=1, $\eta$=0.5, E=10 | E=10 | 1 |
| $\epsilon = 0.3$ | R=3, $\eta$=0.5, E=5 | R=3, $\eta$=0.2, E=10 | C=2, R=1, $\eta$=0.2, E=5 | E=5 | 1 |
| no DP | R=100, $\eta$=2.0, E=3 | R=100, $\eta$=0.2, E=3 | C=2, R=30, $\eta$=0.5, E=3 | E=5 | 30 |
| regional* | R=30, $\eta$=0.5, E=10 | R=3, $\eta$=0.2, E=10 | C=2, R=5, $\eta$=0.2, E=10 | E=10 | 3 |
| wrong anchor* | R=100, $\eta$=0.2, E=5 | R=30, $\eta$=0.2, E=10 | C=4, R=1, $\eta$=2.0, E=30 | E=30 | 10 |
| $\tau^2 = 0.1$, $\epsilon = 0.5$* | R=5, $\eta$=0.5, E=5 | R=3, $\eta$=0.2, E=10 | C=2, R=2, $\eta$=0.2, E=10 | E=10 | 3 |
| $K = 48$* | R=3, $\eta$=2.0, E=3 | R=3, $\eta$=0.2, E=5 | C=2, R=10, $\eta$=0.2, E=5 | E=5 | 10 |

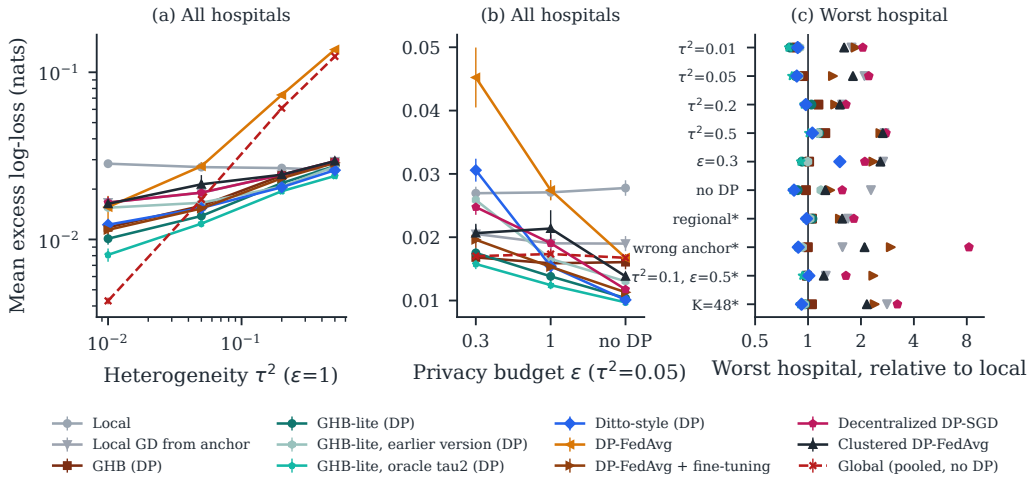



Fig. S14. Simulation 15. (a) Mean excess log-loss against heterogeneity at $\epsilon = 1$. (b) Against the privacy budget at $\tau^2 = 0.05$. (c) The worst hospital's error relative to its own local estimate; right of 1, some hospital is worse off than alone.

$\tau^2 = 0.05$; an anchor wrong by $\mathcal{N}(0, I)$; and a case-mix shift of one standard deviation. In every replicate, every variant uses the same data, the same public sample and the same standard-normal noise draw, scaled to its own noise level, so that differences between variants are paired. Each variant switches one component of GHB-lite (`simulations/ghb_variants.py`, which reproduces the reported estimator to $10^{-10}$ with its default settings): the release (the sufficient statistic $X^\top y$,

Table S17. Simulation 16, part B: the Gaussian model. Mean relative risk $R_k/v_k$ over the 24 hospitals of Simulation 2 (20,000 replicates per setting). FB: $\tau$ integrated out with the hospital's own evidence (reported); Mean, DL: plug-in posterior mean and DerSimonian-Laird [DerSimonian and Laird 1986]; No ev.: components without own evidence; Equal: equal weights; Exact: exact averaging; $T = 10$: gossip with 10 rounds; Oracle: $\tau^2$ known.

| $\tau^2$ | $\epsilon$ | FB | Mean | DL | No ev. | Equal | Exact | $T = 10$ | Oracle |
|---|---|---|---|---|---|---|---|---|---|
| 0.001 | 0.3 | 0.358 | 0.423 | 0.342 | 0.338 | 0.657 | 0.358 | 0.398 | 0.259 |
| 0.02 | 0.3 | 0.741 | 0.735 | 0.908 | 0.771 | 0.786 | 0.741 | 0.752 | 0.689 |
| 0.2 | 0.3 | 0.955 | 0.958 | 0.970 | 0.971 | 0.955 | 0.955 | 0.956 | 0.944 |
| 0.001 | 1 | 0.214 | 0.230 | 0.211 | 0.202 | 0.310 | 0.214 | 0.236 | 0.168 |
| 0.02 | 1 | 0.710 | 0.709 | 0.720 | 0.717 | 0.716 | 0.710 | 0.716 | 0.674 |
| 0.2 | 1 | 0.949 | 0.949 | 0.954 | 0.949 | 0.949 | 0.949 | 0.950 | 0.943 |
| 0.001 | $\infty$ | 0.189 | 0.195 | 0.186 | 0.178 | 0.209 | 0.188 | 0.206 | 0.153 |
| 0.02 | $\infty$ | 0.696 | 0.695 | 0.697 | 0.696 | 0.697 | 0.696 | 0.701 | 0.671 |
| 0.2 | $\infty$ | 0.947 | 0.946 | 0.950 | 0.947 | 0.947 | 0.947 | 0.948 | 0.942 |

or a ridge fit with $\lambda = 0.2$ centred on zero and undone at the leave-one-out point, instead of the score at the anchor); the second linearization; the rule for $\tau$ (plug-in posterior mean, posterior median or DerSimonian-Laird [DerSimonian and Laird 1986], or the posterior quantiles without weighting by the hospital's own evidence); the conflict check; the anchor; the default prior; inverse-variance weights in the network mean; gossip against exact averaging; and zero-concentrated against classical calibration of the noise. GHB-lite with the true $\tau^2$, the earlier version of GHB-lite and local estimation are references. Table 6 of the paper gives the mean excess log-loss of each variant relative to GHB-lite over 80 replicates per setting. Part B repeats the ablation of the rule for $\tau$, the weights and the averaging in the Gaussian model of Simulation 2 over 20,000 replicates per setting (Table S17).

*Simulation 17: sensitivity.* Part A varies each setting of GHB-lite, the others at their reported values, in three base settings of the logistic world ($\tau^2 = 0.05$ at $\epsilon = 1$ and 0.3; $\tau^2 = 0.2$ at $\epsilon = 1$), with every value computed on the same data and releases (40 replicates): the number of posterior quantiles of $\tau$, $Q \in \{1, 3, 6, 12, 24\}$ ($Q = 1$ is the plug-in posterior median); the half-Cauchy scale of the prior on $\tau$, $\{0.1, 0.25, 0.5, 1, 2.5\}$; the standard deviation of the default prior, $\{1, 2.5, 5, 10\}$ or none; the conflict threshold $z \in \{2, 2.5, 3, 4, 6\}$; the stated standard deviation of the anchor, $\{0.1, 0.3, 0.6, 1, 2\}$, whose true error is 0.3; the number of gossip rounds, $T \in \{5, 10, 20, 40, 60, 120\}$; and the size of the public sample for the moments, {2,000, 5,000, 20,000, 100,000}. Ranges of the ratio to the reported settings, over all values and base settings: $Q$ 0.98 to 1.06; scale 0.90 to 1.17; default prior 0.90 to 1.03; threshold 1.00 to 1.04; stated anchor uncertainty 1.00 to 1.41; rounds 1.00 to 1.18; public sample 1.00 to 1.00. Part B varies, at $\tau^2 = 0.05$ and $\epsilon = 1$, the number of hospitals ($K \in \{6, 12, 24, 48, 96\}$, sizes drawn as in Simulation 2), the median hospital size ($\{25, 50, 100, 200, 400\}$), the budget ($\epsilon \in \{0.1, 0.2, 0.3, 0.5, 1, 2, 5, \infty\}$, with zero-concentrated calibration, which is valid for every $\epsilon$), $\delta \in \{10^{-3}, 10^{-5}, 10^{-7}, 10^{-9}\}$ and the four topologies of Figure 2 at $K = 24$, each with its own seeds and 40 replicates. GHB-lite's error relative to local estimation ranges from 0.40 to 0.69, and its worst hospital stays within a factor 1.03 of its local error. Figures 11 and 12 of the paper show every point, and Tables 7 and 8 give the values.

*Simulation 18: group fairness.* The logistic world of Simulation 7 ($\epsilon = 1, \delta = 10^{-5}$, 24 hospitals, ring with two-hop links, $T = 60$) with features $x = (1, S, z_1, z_2, z_3)$: $S \in \{0, 1\}$ is a protected attribute ($S = 1$ the minority group) and $z \sim \mathcal{N}(0, I_3)$ is clipped to $\|z\| \le \sqrt{3}$, so that $\|x\| \le \sqrt{5}$ and the sensitivity of the release is unchanged. Hospital $k$'s minority share is $\pi_k \sim \text{Beta}(6\bar{\pi}, 6(1-\bar{\pi}))$, clipped to $[0.02, 0.98]$, with $\bar{\pi} = 0.3$ (0.1 in the rare-minority scenario); the public feature distribution used for the inversion has share $\bar{\pi}$, so hospitals whose share differs are under case-mix shift. $\theta_k = \mu + u_k$ with $\mu = (-0.5, \mu_S, 0.8, -0.6, 0.4)$, $u_k \sim \mathcal{N}(0, \text{diag}(\tau^2, \tau_S^2, \tau^2, \tau^2, \tau^2))$, $\tau^2 = 0.05$, and $(\mu_S, \tau_S^2) = (0.5, 0.05)$ (shared), $(0.5, 0.5)$ (varies), $(0.5, 0.05)$ with $\bar{\pi} = 0.1$ (rare) or $(0, 0.5)$ (opposite); the anchor is $\mu$ plus $\mathcal{N}(0, 0.3^2 I)$. *GHB-lite, S local* replaces, in every mixture component of each hospital's prior, the network estimate of the coefficient of $S$ and its variance by the local prior $\mathcal{N}(\mu_{0,S}, 1)$. FAIR-FATE follows Algorithm 1 of [Salazar et al. 2023] with the settings in the caption of Table 9 of the paper; the fairness of a model on the server's validation set is its SP, EO or EQO ratio at threshold 0.5, in the form $\min(a, b)/\max(a, b)$, and EQO averages the ratios of true and false positive rates. The validation set holds a quarter as many patients as each hospital's training data, drawn from that hospital's distribution, so a fifth of all patients, as in the paper. Each method is evaluated on 4,000 new patients per hospital, drawn from the hospital's own distribution; the fairness ratios are computed on the pooled test patients of all hospitals, each scored by the model its hospital uses. Results over 100 replicates per scenario are in Table 9 and Figure 13 of the paper. The code is `simulations/exp18_fairness.py`.

## S2 REAL DATA: TCGA DEVELOPMENT, ABLATION AND CONFIRMATION

*Data.* Open-access clinical data of The Cancer Genome Atlas [Cancer Genome Atlas Research Network et al. 2013; Grossman et al. 2016] from the NCI Genomic Data Commons (https://api.gdc.cancer.gov/cases, script `simulations/fetch_tcga.py`, files in `data/tcga_v3/` and `data/tcga_v4/`), downloaded with the fields vital status, days to death, age at index, sex at birth, days to last follow-up, AJCC pathologic stage, T and N, FIGO stage, prior malignancy, primary-diagnosis flag and the tissue source site. A hospital is a tissue source site (by name) with at least five eligible patients. A patient is eligible with a known age, sex and pathologic stage (AJCC, or FIGO where AJCC is absent), and a two-year outcome: death within 730 days, or known survival beyond it.

*Model.* $x = (1, a, \text{male}, \text{stage III}, \text{stage IV}, \text{T3 or T4}, \text{N1 to N3}, \text{prior malignancy})$ with $a = (\text{age} - 62)/25$ clipped to $[-1, 1]$, so $\|x\| \le \sqrt{8}$ without fitting anything to the data. Stage letters and digits are dropped (IIIC1 is III).

*Sets.* Development: LUSC, HNSC, BLCA, STAD, LIHC, KIRC (the first set), and LUAD, COAD, READ, KIRP, PAAD, ESCA (the second set, which became development data once the pre-registered confirmation of the extensions of GHB-lite had run on it). Confirmation: every further TCGA project with at least four hospitals and at least fifteen deaths, fixed from counts alone before any model was fitted: BRCA, MESO, UCEC, OV and UCS. Of the other projects with downloadable data, ACC, PRAD and CESC have no or too few eligible hospitals, and THCA, KICH and TGCT too few deaths.

*Protocol.* Ten repetitions of five-fold cross-validation, stratified by outcome within each hospital, with the networks of a set fitted jointly so that GHB can borrow across them; every other method is fitted per network. Budgets $\epsilon \in \{1, 10\}$, $\delta = 10^{-5}$, zero-concentrated accounting. Errors are pooled held-out log-loss over repetitions and hospitals, relative to local estimation under $\mathcal{N}(0, 2.5^2)$. The baselines and their grids are in Table S18. Script `simulations/real_v3.py` (sets `dev`, `fresh`, `confirm`), with checkpoints per repetition.

Table S18. TCGA: the literature methods, the size of each grid, and the configuration with the lowest mean error over the twelve development networks, frozen before the confirmation (`dev/v4_realistic_configs.json`). Every method spends the same budget, with private gradients under zero-concentrated accounting.

| Method | Grid | $\epsilon = 1$ | $\epsilon = 10$ |
|---|---|---|---|
| DP-FedAvg + FT [McMahan et al. 2018] | 36 | rounds $R$=1, step $\eta$=0.2, fine-tuning steps=5 | rounds $R$=10, step $\eta$=0.2, fine-tuning steps=5 |
| DP-FedAvg + FT, sec. agg. [Bonawitz et al. 2017; McMahan et al. 2018] | 18 | rounds $R$=3, step $\eta$=0.5, fine-tuning steps=5 | rounds $R$=10, step $\eta$=2.0, fine-tuning steps=5 |
| APFL [Deng et al. 2020] | 18 | rounds $R$=3, step $\eta$=0.5, mixing $\alpha$=0.75 | rounds $R$=10, step $\eta$=0.5, mixing $\alpha$=0.5 |
| Ditto [Li et al. 2021] | 18 | rounds $R$=3, step $\eta$=0.5, proximal $\lambda$=1.0 | rounds $R$=3, step $\eta$=0.5, proximal $\lambda$=1.0 |
| pFedMe [T. Dinh et al. 2020] | 12 | rounds $R$=3, step $\eta$=2.0, proximal $\lambda$=10.0 | rounds $R$=3, step $\eta$=2.0, proximal $\lambda$=10.0 |
| Clustered DP-FedAvg [Ghosh et al. 2020; McMahan et al. 2018] | 54 | clusters=2, rounds $R$=1, step $\eta$=0.2, fine-tuning steps=5 | clusters=2, rounds $R$=1, step $\eta$=0.5, fine-tuning steps=5 |

Table S19. TCGA development: twelve networks, ten repetitions; held-out log-loss relative to local estimation, mean over the networks. GHB as frozen; GHB-lite with the evidence-weighted default prior, alone and with the four extensions of the third pre-registration. Literature methods: tuned on the same twelve networks (development tuning, in sample, which favours them) and on each network's own held-out folds (oracle).

| | $\epsilon = 1$ | | $\epsilon = 10$ | |
|---|---|---|---|---|
| Method | Development | Oracle | Development | Oracle |
| GHB, no tuning | **0.926** | 0.926 | **0.904** | 0.904 |
| GHB-lite (evidence prior), no tuning | 0.927 | 0.927 | 0.941 | 0.941 |
| GHB-lite with safety | 0.927 | 0.927 | 0.938 | 0.938 |
| GHB-lite with learned scale | 0.928 | 0.928 | 1.002 | 1.002 |
| GHB-lite with correlated | 0.927 | 0.927 | 0.940 | 0.940 |
| GHB-lite with cross-cancer | 0.928 | 0.928 | 0.934 | 0.934 |
| GHB-lite with all four additions (v3) | 0.928 | 0.928 | 0.981 | 0.981 |
| Local, evidence prior | 0.927 | 0.927 | 0.927 | 0.927 |
| DP-FedAvg + FT [McMahan et al. 2018] | 0.934 | **0.920** | 0.927 | 0.910 |
| DP-FedAvg + FT, sec. agg. [Bonawitz et al. 2017; McMahan et al. 2018] | 0.940 | 0.930 | 0.914 | **0.897** |
| APFL [Deng et al. 2020] | 0.950 | 0.945 | 0.925 | 0.909 |
| Ditto [Li et al. 2021] | 1.012 | 0.991 | 0.916 | 0.908 |
| pFedMe [T. Dinh et al. 2020] | 1.078 | 1.057 | 0.913 | 0.903 |
| Clustered DP-FedAvg [Ghosh et al. 2020; McMahan et al. 2018] | 0.944 | 0.929 | 0.931 | 0.918 |

*Development of GHB.* GHB was selected among about fifteen variants on the development networks (`simulations/dev/v4_dev.py`, with the same folds as `real_v3.py`, so that the stored baselines are compared fold for fold). A first version iterated the joint releases as Newton's method with the information matrix released at every iterate and no propagation of the privacy noise; its network estimate was dominated by noise at $\epsilon = 10$ in networks of a few hundred patients, because a Newton step divides the noise by the information of rare features. Releasing the information once, at the origin, propagating the privacy covariance into each hospital's prior, and borrowing across networks fixed this (Table S20). One Newton step was better than two, which spend half the budget per step. Learning $\tau$ from per-hospital releases at half the budget did not help, and adding institution effects across networks changed nothing to three decimals. Removing a hospital's own data from the network estimate makes almost no difference on average (at $\epsilon = 10$ the variant without it is slightly more accurate, 5 of 12 networks); it is kept because Proposition 4.1 requires an estimate independent of the hospital, and with few hospitals per network a hospital's own share of

Table S20. TCGA development: ablation of GHB, one change at a time from the frozen configuration, on the twelve development networks (ten repetitions, the same privacy noise for every variant). Mean relative held-out log-loss, and the number of networks in which the frozen GHB is more accurate than the variant.

| | $\epsilon = 1$ | | $\epsilon = 10$ | |
|---|---|---|---|---|
| Variant | Mean | GHB better | Mean | GHB better |
| GHB (as frozen) | **0.927** | | **0.907** | |
| without borrowing across networks | 0.958 | 12/12 | 0.920 | 10/12 |
| without the non-borrowing components | 0.928 | 6/12 | 0.908 | 6/12 |
| without propagating the privacy covariance | 0.962 | 12/12 | 0.914 | 12/12 |
| without removing the hospital's own data | 0.927 | 9/12 | **0.904** | 5/12 |
| two rounds of joint releases | 0.927 | 6/12 | 0.913 | 8/12 |
| $\tau$ from per-hospital releases (half the budget) | **0.926** | 7/12 | 0.911 | 8/12 |
| the same, with institution effects across networks | **0.926** | 7/12 | 0.911 | 8/12 |

Table S21. TCGA confirmation: every literature method in the oracle arm, tuned on each network's own held-out folds; relative held-out log-loss. Compare with Table 10 of the paper.

| Method | BRCA | MESO | OV | UCEC | UCS | Mean |
|---|---|---|---|---|---|---|
| $\epsilon = 1$ | | | | | | |
| DP-FedAvg + FT [McMahan et al. 2018] | **0.895** | **0.914** | 1.003 | 0.964 | **0.917** | **0.939** |
| DP-FedAvg + FT, sec. agg. [Bonawitz et al. 2017; McMahan et al. 2018] | 0.911 | 1.027 | 0.997 | **0.962** | 1.019 | 0.983 |
| APFL [Deng et al. 2020] | 0.918 | 0.987 | 1.008 | 0.997 | 0.986 | 0.979 |
| Ditto [Li et al. 2021] | 0.967 | 1.480 | 0.978 | 0.997 | 1.556 | 1.196 |
| pFedMe [T. Dinh et al. 2020] | 0.979 | 1.575 | **0.974** | 1.013 | 1.661 | 1.241 |
| Clustered DP-FedAvg [Ghosh et al. 2020; McMahan et al. 2018] | 0.919 | 0.989 | 1.007 | 0.971 | 0.965 | 0.970 |
| $\epsilon = 10$ | | | | | | |
| DP-FedAvg + FT [McMahan et al. 2018] | 0.888 | **0.877** | 0.981 | 0.964 | **0.864** | 0.915 |
| DP-FedAvg + FT, sec. agg. [Bonawitz et al. 2017; McMahan et al. 2018] | 0.880 | 0.905 | 0.970 | 0.925 | 0.879 | 0.912 |
| APFL [Deng et al. 2020] | 0.882 | 0.888 | 0.989 | 0.957 | 0.878 | 0.919 |
| Ditto [Li et al. 2021] | 0.878 | 0.893 | 0.973 | 0.946 | 0.866 | 0.911 |
| pFedMe [T. Dinh et al. 2020] | **0.845** | 0.900 | **0.962** | **0.900** | 0.885 | **0.898** |
| Clustered DP-FedAvg [Ghosh et al. 2020; McMahan et al. 2018] | 0.892 | 0.922 | 0.996 | 0.965 | 0.876 | 0.930 |

the network estimate is not negligible. The non-borrowing components change the mean by 0.001 and are kept as the safeguard of Remark 4.4.

*Tuning GHB.* After the confirmation, and outside the pre-registration, GHB was tuned like the literature methods (Table 11 of the paper; `simulations/dev/v4_tune.py` and `v4_tune_score.py`). The grid has 36 configurations: the half-Cauchy scale of $\tau$ in $\{0.1, 0.25, 0.5, 1\}$, the total weight of the non-borrowing components relative to the borrowing ones in $\{0, \frac{1}{3}, 1\}$, and the share of the budget given to the information release in $\{\frac{1}{4}, \frac{1}{2}, \frac{3}{4}\}$; the frozen configuration is $(0.5, \frac{1}{3}, \frac{1}{2})$. All configurations share the folds of `real_v3.py` and, within a repetition and fold, the same privacy noise. Tuned on the twelve development networks, the chosen configurations were $(1, 1, \frac{1}{4})$ at $\epsilon = 1$ and $(1, 0, \frac{1}{2})$ at $\epsilon = 10$; on the development networks they improve on the frozen configuration by 0.001 in mean relative error.

## S3 PRE-REGISTRATIONS AND THEIR OUTCOMES

Every change to GHB made after seeing real data was tested under a pre-registration committed to the repository before its confirmation run (`simulations/dev/PREREG_*.md`). The rule and the criteria were fixed in advance, and each outcome was appended afterwards, as it came.

(1) **Evidence-weighted default prior** (`PREREG_prior.md`). GHB-lite integrates the scale of the default prior over $\{0.5, 1, 2.5\}$ by each hospital's evidence. Criteria: improve on GHB-lite in

Table S22. Eight-network benchmark, $\epsilon = 1$, leave-one-network-out arm: held-out log-loss relative to local estimation, per network.

| | Medicare | LUSC | HNSC | BLCA | STAD | LIHC | KIRC |
|---|---|---|---|---|---|---|---|
| Hospitals $K$ | 43 | 17 | 12 | 13 | 9 | 7 | 7 |
| **GHB**, no tuning | 0.961 | 0.964 | 0.943 | 0.944 | 0.968 | 0.926 | 1.023 |
| GHB-lite, no tuning | 0.998 | 1.000 | 1.000 | 1.002 | 1.006 | 1.000 | 0.996 |
| GHB-lite, evidence prior, no tuning | 0.966 | 0.959 | 0.941 | 0.933 | 0.966 | **0.922** | 1.005 |
| GHB-lite with secure aggregation, no tuning | 0.965 | 1.007 | 0.970 | 0.955 | 0.984 | 0.962 | 1.033 |
| GHB-lite, tuned prior | 0.961 | **0.955** | 0.941 | **0.928** | **0.959** | 0.928 | 0.999 |
| Local, tuned prior[‡] | 0.961 | 0.955 | 0.940 | 0.927 | 0.957 | 0.928 | 1.002 |
| DP-FedAvg [McMahan et al. 2018] | 1.040 | 1.074 | 0.967 | 1.033 | 1.045 | 1.160 | 1.462 |
| DP-FedAvg + FT [McMahan et al. 2018] | 0.960 | 0.962 | 0.981 | 1.002 | 0.964 | 1.003 | 1.089 |
| DP-FedAvg + FT, sec. agg. [McMahan et al. 2018] | 0.957 | 0.961 | **0.931** | 1.000 | 0.963 | 1.005 | 1.078 |
| Decentralized DP-SGD [Lian et al. 2017] | 0.971 | 0.978 | 0.992 | 1.011 | 0.990 | 1.020 | 1.099 |
| Clustered DP-FedAvg [Ghosh et al. 2020; McMahan et al. 2018] | 0.960 | 0.964 | 0.984 | 1.006 | 0.966 | 1.005 | 1.092 |
| Ditto-style (one-shot) [Li et al. 2021] | 1.065 | 1.310 | 1.292 | 1.694 | 1.328 | 1.416 | 1.224 |
| Ditto [Li et al. 2021] | 0.961 | 0.962 | 0.947 | 0.949 | 0.964 | 0.953 | 1.001 |
| pFedMe [T. Dinh et al. 2020] | **0.953** | 0.991 | 0.943 | 0.954 | 0.966 | 1.012 | **0.981** |
| APFL [Deng et al. 2020] | 0.957 | 0.966 | 0.961 | 0.976 | 0.986 | 0.971 | 1.101 |
| Per-FedAvg [Fallah et al. 2020] | 1.026 | 1.122 | 1.033 | 1.162 | 1.116 | 1.275 | 1.423 |
| FedRep [Collins et al. 2021] | 0.981 | 0.975 | 1.007 | 1.138 | 1.074 | 1.071 | 1.075 |
| FedProx [Li et al. 2020] | 0.970 | 0.997 | 1.002 | 1.047 | 1.018 | 1.044 | 1.010 |

at least five of the seven networks of the eight-network benchmark, and raise no network's worst-hospital ratio by more than 0.1. Outcome: improved in 6 networks, but raised the worst ratio in the liver network from 1.56 to 1.86, through a hospital whose 22 patients all died. Not adopted as the default of GHB-lite; adopted inside GHB, whose non-borrowing components cover that case.

(2) **GHB-SA** (`PREREG_sa.md`). Secure aggregation of per-hospital clipped influences, with the network still recovered under a public feature distribution. Criteria: improve on the evidence-weighted GHB-lite in at least five of the seven networks at $\epsilon = 1$, with no worst-hospital ratio higher by more than 0.1. Outcome: improved in 1 of seven. Not adopted. It was the first step toward joint releases; its failure pointed to the public feature distribution as the remaining problem.

(3) **Four extensions of GHB-lite** (`PREREG_v3.md`): no-borrowing components, a network-learned prior scale, correlated heterogeneity, and borrowing across cancer types; the configuration with borrowing across cancer types was locked. All four predictions held on six then-unseen networks, but at $\epsilon = 10$ every variant was worse than local estimation with the same prior, and at $\epsilon = 1$ none was better. The diagnosis, bias and noise of per-hospital releases, led to GHB.

(4) **GHB** (`PREREG_v4.md`). Three predictions on five unseen networks, all held (Section 7.1).

## S4 REAL DATA: EIGHT SMALLER NETWORKS

These tables give every network's result behind Table 12 of the paper, in the leave-one-network-out arm: every literature method uses the configuration with the lowest mean error over the other networks. Medicare is Arizona Medicare (43 hospitals); the TCGA networks use the five-coefficient head of the paper and are fitted one at a time. Values are held-out log-loss relative to local estimation; bold marks the lowest per network among methods that use other hospitals.

## S5 REAL DATA: UCI HEART DISEASE

*Data.* The processed Cleveland, Hungarian, Switzerland and VA Long Beach files of the UCI Heart Disease data [Detrano et al. 1989] (CC BY 4.0), as distributed by the UCI Machine Learning

Table S23. Eight-network benchmark, $\epsilon = 10$, leave-one-network-out arm: held-out log-loss relative to local estimation, per network.

| | Medicare | LUSC | HNSC | BLCA | STAD | LIHC | KIRC |
|---|---|---|---|---|---|---|---|
| Hospitals $K$ | 43 | 17 | 12 | 13 | 9 | 7 | 7 |
| **GHB**, no tuning | 0.951 | 0.957 | 0.932 | 0.917 | 0.949 | **0.921** | 0.980 |
| GHB-lite, no tuning | 0.964 | 0.983 | 0.953 | 0.954 | 0.945 | 0.965 | 0.987 |
| GHB-lite, evidence prior, no tuning | 0.954 | 0.968 | 0.936 | 0.923 | 0.949 | 0.923 | 0.989 |
| GHB-lite with secure aggregation, no tuning | 0.955 | 0.974 | 0.935 | 0.925 | 0.947 | 0.930 | 0.983 |
| GHB-lite, tuned prior | 0.956 | 0.968 | 0.938 | 0.914 | 0.948 | 0.933 | 0.994 |
| Local, tuned prior[‡] | 0.961 | 0.955 | 0.940 | 0.927 | 0.957 | 0.928 | 1.002 |
| DP-FedAvg [McMahan et al. 2018] | 0.979 | 1.041 | 0.957 | 0.993 | 1.024 | 1.151 | 1.161 |
| DP-FedAvg + FT [McMahan et al. 2018] | 0.948 | 0.963 | 0.932 | 0.914 | 0.956 | 0.942 | 1.012 |
| DP-FedAvg + FT, sec. agg. [McMahan et al. 2018] | **0.942** | **0.946** | 0.933 | **0.903** | **0.926** | 0.981 | 0.988 |
| Decentralized DP-SGD [Lian et al. 2017] | 0.956 | 0.962 | 0.932 | 0.919 | 0.962 | 0.946 | 1.071 |
| Clustered DP-FedAvg [Ghosh et al. 2020; McMahan et al. 2018] | 0.957 | 0.962 | **0.931** | 0.926 | 0.962 | 0.962 | 1.070 |
| Ditto-style (one-shot) [Li et al. 2021] | 0.985 | 0.995 | 0.977 | 1.193 | 0.975 | 0.996 | 0.984 |
| Ditto [Li et al. 2021] | 0.959 | 0.954 | 0.939 | 0.917 | 0.951 | 0.936 | 0.988 |
| pFedMe [T. Dinh et al. 2020] | 0.956 | 0.950 | 0.938 | 0.909 | 0.934 | 0.951 | **0.959** |
| APFL [Deng et al. 2020] | 0.951 | 0.962 | 0.938 | 0.947 | 0.962 | 0.955 | 1.026 |
| Per-FedAvg [Fallah et al. 2020] | 0.982 | 1.013 | 0.943 | 0.964 | 1.008 | 1.071 | 1.197 |
| FedRep [Collins et al. 2021] | 0.977 | 0.956 | 0.993 | 1.097 | 1.039 | 1.049 | 1.051 |
| FedProx [Li et al. 2020] | 0.951 | 0.961 | 0.931 | 0.922 | 0.959 | 0.935 | 1.046 |

Repository; they are in `data/heart_disease/` of the repository. The outcome is angiographic coronary disease (num $> 0$). Sizes and prevalences: Cleveland 303 (46%), Budapest 294 (36%), Zurich and Basel 123 (93%), Long Beach 200 (75%). Age, sex and chest-pain type are complete at every site; the exercise-test variables are missing for about a quarter of the Long Beach patients, and cholesterol is recorded as zero at the Swiss sites, so they are not used.

*Model and privacy.* $x = (1, a, \text{male}, \text{asymptomatic}, \text{anginal})$ with $a = (\text{age} - 55)/25$ clipped to $[-1, 1]$; non-anginal pain is the reference, and anginal combines typical and atypical angina. Hence $\|x\| \le 2 \le \sqrt{5}$, and the release (the mean score at 0, $\epsilon \in \{1, 5\}$, $\delta = 10^{-5}$, zero-concentrated calibration) has the sensitivity used throughout. GHB-lite is `ghb_logit.py` without an anchor: the conflict check is off, the second linearization point is the leave-one-out network estimate combined with the default prior $\mathcal{N}(0, 2.5^2)$ in place of an anchor (without it, the linearization point of a hospital with 20 patients can reach the clipping bound and make the public information matrix singular), and the final fit uses only the network components and the default prior. Gossip runs for 60 rounds on the ring of four hospitals. The public feature distribution is age $\mathcal{N}(55, 9^2)$, male with probability 0.75 and chest pain asymptomatic, anginal or non-anginal with probabilities 0.5, 0.3 and 0.2, independently: a nominal referral population, not estimated from the data. With the public information at 0, the standard deviation of the privacy noise on the parameter scale is, at the median coefficient, 26 times the sampling standard error of a hospital of 40 patients at $\epsilon = 1$, 5.6 times at $\epsilon = 5$, and 9.5 and 2.0 times for a hospital of 300 patients.

*Evaluation.* Twenty repeats of stratified five-fold cross-validation within each hospital, with the folds of the four hospitals aligned so that every split has one training set and one test set per hospital. The training set is subsampled, stratified by outcome, to at most 20, 40 or 80 patients, or kept whole. All methods are fitted on the training sets and scored on each hospital's own test fold by log-loss, Brier score and AUC. Baselines (`simulations/literature.py`): DP-FedAvg [McMahan et al. 2018] from 0 with $R \in \{1, 3, 10, 30\}$ rounds, step $\eta \in \{0.2, 0.5, 2\}$ and $E \in \{0, 1, 5, 30\}$ full-batch fine-tuning steps of size 1; the Ditto-style [Li et al. 2021] model, a MAP fit with prior $\mathcal{N}(\bar{\theta}, \lambda^{-1} I)$ around the size-weighted mean $\bar{\theta}$ of GHB-lite's inverted releases at $\epsilon = 1$, $\lambda \in \{0.01, 0.1, 0.3, 1, 3, 10, 30, 100\}$; and local gradient descent from 0 with step 1 and $E \in \{1, 2, 3, 5, 10, 30, 100\}$ steps. Each is tuned by

Table S24. UCI Heart Disease: hyperparameters of the baselines chosen on the held-out folds, by the maximum number of training patients per hospital.

| Training patients | Local GD, tuned | DP-FedAvg [McMahan et al. 2018] + FT ($\epsilon = 1$) | Ditto-style [Li et al. 2021] ($\epsilon = 1$) |
|---|---|---|---|
| 20 | E=30 | R=1, $\eta$=0.2, E=30 | $\lambda$=0.01 |
| 40 | E=30 | R=3, $\eta$=0.2, E=30 | $\lambda$=0.01 |
| 80 | E=100 | R=10, $\eta$=0.2, E=30 | $\lambda$=0.01 |
| all | E=100 | R=3, $\eta$=2.0, E=30 | $\lambda$=0.1 |

Table S25. UCI Heart Disease: held-out log-loss, Brier score and AUC of every method at each hospital (Cle: Cleveland; Bud: Budapest; Swi: Zurich and Basel; VA: Long Beach), mean over 100 splits. †: not private.

| | Log-loss | | | | Brier score | | | | AUC | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Method | Cle | Bud | Swi | VA | Cle | Bud | Swi | VA | Cle | Bud | Swi | VA |
| *At most 40 training patients* | | | | | | | | | | | | |
| Local | 0.533 | 0.514 | 0.232 | 0.563 | **0.175** | 0.165 | 0.063 | 0.186 | 0.820 | 0.814 | 0.750 | 0.644 |
| GHB-lite, $\epsilon = 1$ | **0.532** | 0.514 | 0.232 | 0.563 | **0.175** | 0.165 | 0.063 | 0.187 | **0.821** | 0.813 | 0.745 | 0.645 |
| GHB-lite, $\epsilon = 5$ | 0.533 | 0.510 | 0.234 | 0.556 | 0.176 | 0.164 | 0.064 | 0.184 | **0.821** | 0.818 | 0.738 | 0.659 |
| DP-FedAvg + FT, $\epsilon = 1$ [McMahan et al. 2018] | 0.542 | **0.501** | **0.217** | **0.541** | 0.179 | **0.163** | **0.058** | **0.179** | 0.819 | **0.825** | **0.776** | 0.666 |
| Ditto-style, $\epsilon = 1$ [Li et al. 2021] | 0.616 | 0.628 | 0.479 | 0.720 | 0.182 | 0.173 | 0.076 | 0.202 | 0.817 | 0.801 | 0.637 | 0.622 |
| Local GD, tuned stopping | 0.542 | **0.501** | **0.217** | 0.542 | 0.179 | **0.163** | **0.058** | 0.180 | 0.819 | **0.825** | 0.766 | **0.668** |
| GHB-lite, no DP† | 0.516 | 0.491 | 0.231 | 0.542 | 0.169 | 0.159 | 0.063 | 0.180 | 0.833 | 0.828 | 0.729 | 0.690 |
| GHB-lite, empirical features† | 0.525 | 0.494 | 0.226 | 0.537 | 0.173 | 0.160 | 0.061 | 0.178 | 0.829 | 0.827 | 0.733 | 0.685 |
| Global† | 0.528 | 0.512 | 0.364 | 0.537 | 0.174 | 0.167 | 0.108 | 0.177 | 0.835 | 0.831 | 0.737 | 0.709 |
| *All training patients* | | | | | | | | | | | | |
| Local | 0.502 | 0.485 | 0.220 | 0.529 | 0.165 | 0.156 | 0.060 | 0.174 | **0.841** | 0.832 | 0.737 | 0.689 |
| GHB-lite, $\epsilon = 1$ | 0.502 | 0.484 | 0.220 | 0.529 | 0.165 | 0.156 | 0.060 | 0.174 | **0.841** | 0.832 | 0.748 | 0.692 |
| GHB-lite, $\epsilon = 5$ | **0.501** | 0.484 | 0.225 | 0.528 | **0.164** | **0.155** | 0.061 | 0.174 | 0.840 | 0.831 | **0.754** | 0.698 |
| DP-FedAvg + FT, $\epsilon = 1$ [McMahan et al. 2018] | 0.525 | 0.487 | **0.211** | **0.524** | 0.172 | 0.158 | **0.057** | **0.173** | 0.826 | 0.834 | 0.752 | **0.706** |
| Ditto-style, $\epsilon = 1$ [Li et al. 2021] | 0.503 | 0.486 | 0.228 | 0.532 | 0.165 | 0.156 | 0.062 | 0.176 | 0.840 | 0.832 | 0.740 | 0.687 |
| Local GD, tuned stopping | 0.504 | **0.483** | 0.213 | 0.526 | 0.165 | 0.156 | 0.058 | **0.173** | 0.838 | **0.836** | 0.725 | 0.697 |
| GHB-lite, no DP† | 0.501 | 0.481 | 0.224 | 0.526 | 0.164 | 0.155 | 0.061 | 0.174 | 0.838 | 0.833 | 0.750 | 0.699 |
| GHB-lite, empirical features† | 0.504 | 0.480 | 0.219 | 0.525 | 0.165 | 0.155 | 0.059 | 0.173 | 0.837 | 0.834 | 0.754 | 0.702 |
| Global† | 0.503 | 0.488 | 0.397 | 0.531 | 0.164 | 0.158 | 0.121 | 0.177 | 0.840 | 0.834 | 0.758 | 0.711 |

its mean held-out log-loss over all splits and hospitals, separately for each training size (Table S24). Local estimation and the pooled model are MAP fits with the default prior $\mathcal{N}(0, 2.5^2)$. The code is `simulations/real_heart.py`; Table S25 gives every hospital's results.

## S6 REVISION OF GHB-LITE IN SIMULATION AND EXTENSIONS TRIED

The first run of Simulation 15 showed that the earlier GHB-lite lost to DP-FedAvg [McMahan et al. 2018] with fine-tuning when hospitals were nearly identical and when the budget was strict. A diagnosis on development seeds (900 to 904, used by no reported simulation) found two causes. First, its plug-in $\tau^2$, the posterior mean under a half-Cauchy prior, is driven by the heavy prior tail when releases are noisy: it averaged 0.8 to 1.1 against a true 0.05 at $\epsilon = 0.3$, so GHB-lite barely borrowed. Second, its ridge releases ($\lambda = 0.2$) have five times the sensitivity of a score, and undoing the ridge to first order at a fixed reference left a bias in the releases of the largest hospitals without privacy (their squared errors were 2.5 to 9 times their stated variances). The revision makes three changes, each following the design of Section 4: a score release at the anchor, inverted by public moment matching at anchor-free or check-gated points (scope (d) of Section 4.2); $\tau$ integrated out over its posterior, weighted by each hospital's own marginal likelihood (Remark 4.4); and the anchor conflict check by default, decided once per hospital on anchor-free information. Table S26 gives the development runs that chose between the alternatives; the scripts are in `simulations/dev/`.

Table S26. Development runs (20 replicates each, development seeds): mean excess log-loss of the earlier GHB-lite, the revised GHB-lite, and local estimation. Settings use the logistic world of Simulation 7; "shift" is a case-mix shift of one standard deviation, "anchor" an anchor wrong by two standard deviations.

| | $\tau^2 = 0.05$ | | | $\tau^2 = 0.5$ | $\tau^2 = 0.05, \epsilon = 1$ | |
|---|---|---|---|---|---|---|
| | $\epsilon = 0.3$ | $\epsilon = 1$ | no DP | $\epsilon = 1$ | shift | anchor |
| Local | 0.0261 | 0.0289 | 0.0275 | 0.0272 | 0.0239 | 0.0581 |
| GHB-lite, earlier version | 0.0243 | 0.0166 | 0.0134 | 0.0268 | 0.0151 | 0.0194 |
| GHB-lite, revised | 0.0158 | 0.0143 | 0.0106 | 0.0269 | 0.0138 | 0.0160 |

*Alternatives rejected on the development runs.* Releasing $X^\top y/n$ instead of the score: calibrated, but about twice the sampling variance for the slopes, because it ignores the hospital's own features. Ridge releases centred on the anchor ($\lambda = 1$ or 0.2) or on zero: competitive when the anchor is right, but two to three times worse with a wrong anchor (0.032 to 0.045 against 0.016). Plug-in alternatives to integrating $\tau$ out: the posterior median or the posterior mean of the precision borrowed more at small $\tau^2$ but became worse than local estimation at $\tau^2 = 0.5$ (up to 0.041 against 0.028). An additional Student-$t$ tail on the integrated prior changed nothing beyond Monte Carlo error. Without a weakly informative default prior, the widest components left tiny hospitals' fits unregularized and logistic separation made some diverge; with a conflict check applied separately per component, a wrong anchor re-entered through the wide components.

*Extensions tried and not adopted.* Three extensions proposed in Section 4.2 and the agenda were developed in the same way (`dev/ext.py`, `ext2.py`, `ext3.py`). (1) *Joint release* by groups of $g$ consecutive hospitals through secure aggregation, each group entering borrowing as one unit whose heterogeneity is diluted by its size weights: groups of three over all hospitals lowered the error by at most 12% (0.0140 against 0.0160 at $\tau^2 = 0.01$, $\epsilon = 0.3$; 0.0125 against 0.0133 at $\tau^2 = 0.05$, $\epsilon = 1$) and never raised it beyond Monte Carlo error. It is worth its cost only where secure aggregation is available anyway. (2) *Two release rounds* under zCDP, the second score evaluated at each hospital's public network estimate: no gain in any setting (for example 0.0176 against 0.0177 at $\epsilon = 0.3$, 0.0138 against 0.0135 at $\epsilon = 1$), because splitting the budget cancels the better linearization point. (3) *A two-level prior* for public region labels (hospital within region within network), with both variance components integrated out: a small gain in a three-region world at $\epsilon = 1$ (0.0158 against 0.0164) but losses elsewhere (0.0158 against 0.0141 without regions, 0.0215 against 0.0175 at $\epsilon = 0.3$): two variance components cannot be estimated reliably from 24 noisy releases.

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